

GAS DETECTION EQUIPMENT



Look to Total Safety for the products, programs, people and services that deliver comprehensive, cost-saving solutions for toxic and hazardous gas detection at your facilities.

Equipment Rentals

Short- and long-term rentals are available on premium equipment at reasonable prices—daily, weekly or custom.

Instrument Management and Maintenance (IMM) Program

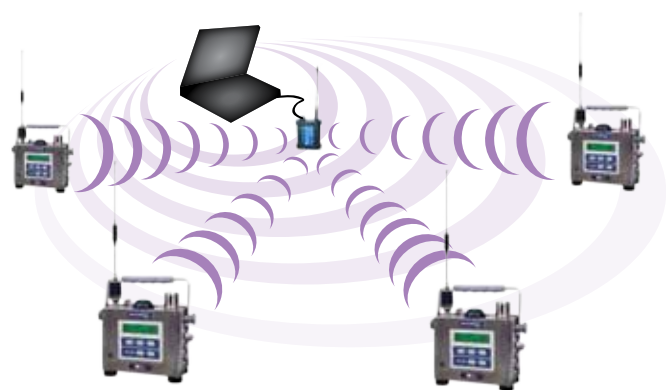
Our program provides asset management of your equipment, designed and executed to ensure the best returns on your equipment investment.

Equipment Service, Compliance and Engineering

Total Safety services all of the equipment that we sell. Also, our technicians have the expertise to repair and maintain all popular respiratory products.

- Trained and certified technicians
- 24/7 service access
- Onsite or in-shop services
- Service agreements built around your needs
- Equipment maintained for optimum performance and regulatory compliance standards anywhere in the world

Wireless Area Monitors – A Total Safety Specialty



Wireless Area Monitoring Network with AreaRAE and RAELink2 Modem connected to a PC.

Gas Detection Equipment

Top-quality gas monitoring equipment that ensures the protection of workers performing a range of applications in non-hazardous and hazardous environments.

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Gas Detection Selection Guide

Product	NH ₃	CO ₂	CO	Cl ₂	ClO ₂	LEL	HCN	H ₂ S	CH ₄	NO	NO ₂	O ₂	O ₃	PH ₃	SO ₂	ETO	VOCs	Pump	Datalog	Page
BW Technologies by Honeywell																				
GasAlert Extreme ⁵	+		+	+	+		+	+		+	+	+	+	+	+	+		+	+	97
GasAlertClip Extreme 2 and 3 ⁵			+					+				+			+					97
GasAlertMax XT II ^M			+			+		+				+						+	+	90
GasAlertMicro 5 IR ^M	+	+	+			+		+				+	+		+			+	+	91
GasAlertMicro 5 ^M	+	+	+	+	+	+	+	+				+	+	+	+		+	+	+	91
GasAlertMicro 5 PID ^M	+		+	+	+	+	+	+				+	+	+	+		+	+	+	91
GasAlertMicroClip XT ^M			+			+		+				+						+	+	91
GasAlertQuattro ^M			+			+		+				+						+	+	90
Rig Rat III ^{A-P}	+		+			+		+				+	+		+				+	102
Detcon																				
Model 10 ^{A-F}	+	+	+	+		+	+	+	+	+	+	+			+		+			120
Model 12 ^{A-F}	+	+	+	+		+	+	+	+	+	+	+			+		+			120
Model 500 ^{A-F}	+	+	+	+		+	+	+	+	+	+	+			+		+	¹	¹	119
Model 600 ^{A-F}	+	+	+	+		+	+	+	+	+	+	+			+		+	¹	¹	119
Model 700 ^{A-F}	+	+	+	+		+	+	+	+	+	+	+			+		+	¹	¹	119
Model 880A/S ^{A-F}	+	+	+	+		+	+	+	+	+	+	+			+		+		¹	121
Model X40 ^{A-F}	+	+	+	+		+	+	+	+	+	+	+			+		+		¹	120
MCX-32 ^{A-F}	+	+	+	+		+	+	+	+	+	+	+			+		+		¹	121
Honeywell																				
MultiPro ^M			+			+		+				+						+	+	95
ToxiPro ^M	+		+	+			+	+				+	+		+				+	100
ISC																				
BM25 Multiguard ^M	+	+	+	+		+	+	+	+	+	+	+			+			+	+	94
GasBadge® Plus ⁵			+					+				+	+		+					100
GasBadge® Pro ⁵	+		+	+			+	+		+	+	+			+				+	99
MX6 iBrid™ ^M	+	+	+	+		+	+	+	+	+	+	+			+		+		+	94
Ventis™ MX4 ^M			+			+		+	+		+	+			+			²	+	93
MSA																				
ALTAIR® 4X ^M			+			+		+	+			+							+	88
ALTAIR® 5X ^M	+	+	+	+		+	+	+	+		+	+			+			+	+	88
ALTAIR® Maintenance-Free ⁵			+					+				+							+	96
ALTAIR® Pro ⁵	+		+	+			+	+				+	+		+				+	96
Orion® ^M			+			+		+				+							+	89
Sirius® PID ^M			+			+		+	+			+					+	+	+	89
Otis																				
GenII Explosion-Proof ^{A-F}	+	+	+	+		+	+	+	+		+	+			+		+			122
GenII Hybrid ^{A-F}	+	+	+	+		+	+	+	+		+	+			+		+			122
GenII Prosafe 4 x 32 ² ^{A-F}	+	+	+	+		+	+	+	+		+	+			+		+			124
GenII Prosafe 4 x 4 ² ^{A-F}	+	+	+	+		+	+	+	+		+	+			+		+			124
GenII Touch-Screen Hybrid ^{A-F}	+	+	+	+		+	+	+	+		+	+			+		+		+	122
Prosafe 4 x 4 ^{A-F}	+	+	+			+		+	+			+			+					123
RAE Systems																				
AreaRAE ^{A-P}	+		+	+		+	+	+		+	+	+			+		+	+	+	103
AreaRAE Steel ^{A-P}	+		+	+		+	+	+		+	+	+			+		+	+	+	103
MeshGuard ^{A-PF}	+		+	+		+		+				+			+				+	104
MiniRAE 3000 ^{PID}																	+	+	+	106
MiniRAE Lite ^{PID}																	+	+		107
MultiRAE Pro ^M	+	+	+	+		+	+	+	+	+	+	+			+		+	+	+	92
ppbRAE 3000 ^{PID}																	+	+	+	107
QRAE II (Diffusion) ^M			+			+		+				+			+				+	92
QRAE II (Pump) ^M			+			+		+				+			+			+	+	92
ToxiRAE 3 ⁵			+					+											+	99
ToxiRAE Pro ⁵	+		+	+		+	+	+		+	+	+			+		+		+	98
UltraRAE 3000 ^{PID}																	+	+	+	106
Pentech																				
Ultra 1000 ^{A-F}	+	+	+	+		+		+	+	+	+	+			+					125
RKI																				
Eagle™ ^M	+	+	+	+		+	+	+	+	+	+	+			+		+	+	+	95

¹ Optional ² Hand Pump ³ Event logging ^M Multigas ⁵ Single Gas ^{PID} Photoionization Detector ^{A-P} Area Monitoring-Portable ^{A-F} Area Monitoring-Fixed ^{A-PF} Area Monitoring-Portable Fixed



PRODUCT SPOTLIGHTS



BW Technologies by Honeywell develops and manufactures gas detection instrumentation to protect personnel and facilities around the world. Reliable, technology-driven instruments provide cost-effective solutions for all your gas detection needs.

Standard on all these BW Multigas Detectors:

- Visual, audible and vibrating alarms **1**
- Full compatibility with BW's MicroDock II test and calibration station **2**
- Full 2-year warranty including sensors
- Multilingual displays

Optional

- Confined Space Kits available for each model **3**



GasAlertMicroClip XT Portable Gas Detector

LEL, O₂, CO or H₂S

Compact and sleek, the GasAlertMicroClip XT provides affordable protection from atmospheric hazards. Features visual compliance with IntelliFlash™, a green LED confidence flash that verifies normal operation to both the user and supervisors from up to 20 ft (6.1 m) away. Simple one-button operation provides ultimate ease of use. Event logging and datalogging are standard.

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**Multigas Monitoring
Made Simple**

GasAlertMax XT II Portable Gas Detector

LEL, O₂, CO or H₂S

The GasAlertMax XT multigas detector offers an integrated sampling pump designed with SmartSample technology to prevent moisture from entering the pump and to save time with reliable pump block detection.

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**Simple Operation
with Integral Pump**

GasAlertQuattro Multi-Gas Detector

LEL, O₂, CO or H₂S

Displays real-time gas concentrations readings for up to four gas hazards. Monitor features visual compliance with IntelliFlash™. Event logging and datalogging are standard. Simple one-button operation provides ultimate ease of use, and the short learning curve significantly reduces the time spent training the user.

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**Flexible, Interchangeable
Power Options for
Extended Runtime**

GasAlertClip Extreme (2 Year Version)

Very Low Cost of Ownership

A very popular and widely used (especially the H₂S version) zero-maintenance gas detector, the GasAlertClip Extreme (2 year version) offers continuous, reliable protection for up to 2 years. No calibration, sensor replacement, battery replacement or battery charging required.

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**2 Years of
Maintenance-
Free Protection**

***Available for purchase or rent.**



PRODUCT SPOTLIGHTS



RAE Systems is a leading global provider of rapidly deployable hazardous gas detection solutions. RAE Systems holds over 18 chemical sensor patents and has developed proprietary photoionization (PID) and wireless area monitoring technologies to meet the needs of leading corporations and many governments worldwide.



For Rapidly Deployable Area Monitoring

AreaRAE Steel Portable Area Monitoring System*

LEL, O₂, H₂S, CO or PID

The AreaRAE is the benchmark for rapidly deployable wireless area monitoring. This one- to five-gas detector features a wireless radio frequency (RF) modem that allows it to communicate and transmit readings on a real-time base with a remotely located base controller. Extremely rugged stainless-steel enclosure stands up to the harshest environments.

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A Highly Advanced PID with Extended Range

UltraRAE 3000 Benzene Monitor*

The UltraRAE 3000 is a highly advanced compound-specific PID with an extended range of 0.05 to 10,000 ppm in VOC mode and 50 ppb to 200 ppm in benzene-specific mode. Ideal for applications from entry pre-screening during refinery and plant maintenance to hazardous material response, marine spill response and refinery downstream monitoring.

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*Wireless Portable Multi-Threat Monitor
for Radiation and Chemical Detection*

MultiRAE Pro

The MultiRAE Pro is the industry's first wireless multi-threat detection tool that combines continuous monitoring capabilities for gamma radiation, volatile organic compounds (VOCs), toxic, combustible gases and oxygen, with Man Down Alarm functionality in one highly portable instrument.

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*The World's First Wireless Personal
Monitor for Toxic Gases and Oxygen*

ToxiRAE Pro Personal Wireless Monitor

The ToxiRAE Pro takes worker protection to the next level by providing safety professionals wireless access to real-time instrument readings and alarm status from any location for better visibility and faster response. Monitor can detect a broad range of gases—from carbon monoxide, hydrogen sulfide and oxygen to ammonia, chlorine and phosgene. Intelligent interchangeable sensors store calibration data, so if you need to monitor one gas one day and another gas the next, swapping the sensors is all it takes.

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**Available for purchase or rent.*



PRODUCT SPOTLIGHTS

INDUSTRIAL SCIENTIFIC

Industrial Scientific Corporation (ISC) is a leader in the design and manufacture of reliable gas detection and monitoring equipment worldwide. ISC's comprehensive product portfolio includes portable multigas and single gas detectors, photoionization detectors (PIDs) and fixed-point gas monitoring systems. All portable detectors are compatible with the DS2 Docking Station™.

MX 6 iBrid™ Multigas Monitor*

1 to 6 Gas Monitor
25 Sensor Options Including PID and IR

The MX 6 iBrid™ color graphic LCD is easily viewable in low to bright light. In addition to millions of possible sensor combinations, you can order an MX 6 iBrid™ configured with either an integral sampling pump (for confined space) or diffusion testing (for personal monitoring).

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*The First Gas Monitor with
a Color Display*

Ventis™ MX4 Multigas Monitor*

1 to 4 Gas Monitor (O₂, H₂S, CO, LEL)

The Ventis MX4 provides reliable monitoring for 1 to 4 gases with a wide range of sensor options. May be configured for diffusion applications or with an integral sampling pump for sample draw applications. Diffusion monitor features a rechargeable lithium-ion extended range battery pack with a 20 hour run-time. Features include LCD readout, peak-hold function, STEL, TWA and datalogging.

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Compact and Lightweight

GasBadge® Plus Personal Single Gas Monitor*

O₂, CO, H₂S, SO₂ or NO₂

Compact and lightweight with a top-mounted sensor, the GasBadge® Plus can be worn comfortably to provide continuous and unobstructed protection even when placed in a shirt pocket. Monitor is extremely durable and resistant to water and abuse in harsh industrial environments. Powered by a non-replaceable lithium-ion battery.

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*Continuous Year of
Maintenance-Free Monitoring*

DS2 Docking Station™*

For a Single Monitor or an Entire Fleet

The DS2 Docking Station™ is ideal for automatic instrument calibration, record-keeping, diagnostics and recharging. It works for a single monitor, or the DS2 can network up to 100 instrument docking stations (IDSs) through Ethernet cable to a docking station server for plant-wide distribution and autonomous operation.

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Networks Up to 100 ISC Gas Monitors

***Available for purchase or rent.**



PRODUCT SPOTLIGHTS

MSA

Established in 1914 and with a strong global presence, MSA is a leading manufacturer of sophisticated gas detectors designed and built with "integrated system" capability that ensures compatibility between individual products.



Customizable Multigas for Many Applications

MSA ALTAIR® 5X Multigas Detector*

LEL, O₂, H₂S, CO, NH₃, Cl₂, or SO₂

The MSA ALTAIR 5 Multigas Detector allows for an industry-first LEL, O₂, CO, H₂S, and SO₂ 60-second span calibration and 10-second bump test time. Other features include monochrome or color display options, interchangeable plug-and-play sensor slots (XCell sensors), MSA's exclusive MotionAlert ("man down" alert), InstantAlert (manual activation of the audible alarm to warn others nearby of danger) and many more great features plus outstanding durability and performance.

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Standalone or Networked Testing

Galaxy® Automated Test System*

LEL, O₂, CO, H₂S or SO₂

The Galaxy® Automated Test System works without the touch of a single button. Simply place the instrument in the test stand and the system performs the test automatically no user intervention is required. The Galaxy System is completely standalone; it does not require a computer or network interface for any mode of operation. Nevertheless, it can be upgraded with a network interface for either wired (Ethernet) or wireless networks.

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***Available for purchase or rent.**

PORTAGAS

Located in Houston, Texas, with a 50,000 sq. ft production facility, PORTAGAS™ is dedicated to providing calibration gases including environmental gases for fugitive emissions and toxic organic mixes and industrial hygiene gases.

PORTAGAS™ Calibration Gases

Calibration Gases and Equipment for Environmental and Industrial Hygiene Applications

PortaGas manufactures all gases to the highest accuracy with traceability to NIST. Cylinders are designed to be light and portable. Each cylinder is labeled with a lot number and expiration date. The stability of each mix is guaranteed.

PortaGreen™ Recyclable Cylinders

PortaGas has developed a recyclable cylinder program utilizing a complete line of reusable, sustainable cylinders. This eliminates the cost and work associated with the current cylinder disposal process. The PortaGreen cylinders can be conveniently returned to PortaGas for reuse using our Strip & ShipSM program. These cylinders are more compact and globally approved.

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Environmental and Industrial Hygiene Mixes



PRODUCT SPOTLIGHTS



Detcon designs and manufactures a wide range of gas detection sensors, control systems and process analyzers. With over 25 years of experience, Detcon is a global company with factory service centers around the world and an expansive inventory of quality products and state-of-the-art solutions. The company's research and development team continues its focus on developing better products and innovative gas detection solutions.

Gas Detection Sensors

Model 700

Model Series 700 is Detcon's newest generation of Intelligent gas detection sensors. Dubbed as "Environmentally Bulletproof," the 700 line includes an electro-polished 316 stainless-steel housing, multi-layered surge protection and 100% encapsulated electronics completely protecting the sensor from water ingress and corrosion. All sensor elements are plug-in field replaceable. Outputs include 4-20 mA and Modbus™ RS-485. Sensor technologies include solid state MOS, catalytic bead, electrochemical, infrared, and photoionization.

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"Environmentally Bulletproof" & Intelligent

OSHA Regulation 29 CFR 1910.146 Confined Space

Many workplaces contain spaces that are considered "confined" because their configurations hinder the activities of employees who must enter, work in and exit them. A confined space has limited or restricted means for entry or exit, and it is not designed for continuous employee occupancy. Confined spaces include, but are not limited to, underground vaults, tanks, storage bins, manholes, pits, silos, process vessels and pipelines. OSHA uses the term "permit-required confined space" (permit space) to describe a confined space that has one or more of the following characteristics: contains or has the potential to contain a hazardous atmosphere; contains a material that has the potential to engulf an entrant; has walls that converge inward or floors that slope downward and taper into a smaller area which could trap or asphyxiate an entrant; or contains any other recognized safety or health hazard, such as unguarded machinery, exposed live wires or heat stress.

Atmospheric testing is required for two distinct purposes:

- Evaluation of the hazards of the permit space
- Verification that acceptable entry conditions for entry into that space exist

(1) **Evaluation testing** – The atmosphere of a confined space should be analyzed using equipment of sufficient sensitivity and specificity to identify and evaluate any hazardous atmospheres that may exist or arise, so that appropriate permit entry procedures can be developed and acceptable entry conditions stipulated for that space. Evaluation and interpretation of these data and development of the entry procedure should be done or reviewed by a technically qualified professional based on evaluation of all serious hazards.

(2) **Verification testing** – The atmosphere of a permit space, which may contain a hazardous atmosphere, should be tested for residues of all contaminants identified by evaluation testing using permit specified

equipment to determine that residual concentrations at the time of testing and entry are within the range of acceptable entry conditions. Results of testing should be recorded on the permit in the space provided adjacent to the stipulated acceptable entry condition.

(3) **Duration of testing** – Measurement of values for each atmospheric parameter should be made for at least the minimum response time of the test instrument specified by the manufacturer.

(4) **Testing stratified atmospheres** – When monitoring for entries involving a descent into atmospheres that may be stratified, the atmospheric envelope should be tested a distance of approximately 4 ft (1.2 m) in the direction of travel and to each side. If a sampling probe is used, the entrant's rate of progress should be slowed to accommodate the sampling speed and detector response.

(5) **Order of testing** – A test for oxygen is performed first because most combustible gas meters are oxygen dependent and will not provide reliable readings in an oxygen deficient atmosphere. Combustible gases are tested for next because the threat of fire or explosion is both more immediate and more life threatening, in most cases, than exposure to toxic gases and vapors. If tests for toxic gases and vapors are necessary, they are performed last.

SOURCE: www.OSHA.gov




MSA

Altair® 5X Multigas Detector



Simultaneously monitors up to six gases, has monochrome or color display options and has interchangeable plug-and-play sensor slots (XCell sensors).

Rugged housing can withstand a 10-ft (3.1 m) drop test, is easy to operate, and is the only instrument in its class with an internally-integrated pump and has a full 3-year warranty on LEL, O₂, H₂S, CO, SO₂ and IR sensors.

MotionAlert feature activates a "man down" alarm if no motion is detected for 30 seconds. InstantAlert feature allows users to manually activate an audible alarm signaling potentially dangerous situations.

Detector also features a 60-second span calibration time for LEL, O₂, CO,

H₂S and SO₂. Sensor response and clear times in under 15 seconds on LEL, O₂, CO, H₂S and SO₂. This detector has typical cost savings of over 50% on calibration gas, replacement sensors and maintenance, and greater signal stability and repeatability under changing or extreme environmental conditions.

Additional features include a 4 year sensor life on LEL, O₂, CO, H₂S, SO₂ and IR sensors. Instrument also has a 3 year sensor life on NH₃ and Cl₂, MSA- exclusive end-of-sensor-life warning, and uses 40% less calibration gas per minute than industry average.

Economy Kit includes monochrome display, integral pump, datalogging, rechargeable Li-ion battery, charger, calibration cap and manual.

Industrial Kit includes Economy kit accessories, as well as a 10-ft sampling line and 1-ft probe.

Deluxe Kit includes Economy and Industrial kit accessories, as well as an upgraded color display.

ECONOMY KITS

Cat. No.	Description	Rental	Purchase Price
MSA10116924	LEL, O ₂ , CO, H ₂ S	YES	\$1,831.00
MSA10116925	LEL, O ₂ , CO, H ₂ S, SO ₂	YES	2,392.00

INDUSTRIAL KITS

Cat. No.	Description	Rental	Purchase Price
MSA10116926	LEL, O ₂ , CO, H ₂ S	YES	\$2,110.00
MSA10116927	LEL, O ₂ , CO, H ₂ S, SO ₂	YES	2,671.00

DELUXE KITS

Cat. No.	Description	Rental	Purchase Price
MSA10116928	LEL, O ₂ , CO, H ₂ S	YES	\$2,389.00
MSA10116929	LEL, O ₂ , CO, H ₂ S, SO ₂	YES	2,931.00

MSA

Altair® 4X Multigas Detector



Detector features a rugged stainless-steel clip, ultra-bright LEDs on both top and bottom of the unit, MSA XCell sensors with protected sensor inlet, 24-hour battery run time and end-of-sensor-life warning. The MotionAlert sensor activates a "man down" alarm when user is motionless for 30 seconds. And with a simple push of a button, the InstantAlert feature enables users to manually activate an audible alarm signaling potentially hazardous situation.

Additional features include large, glove-friendly buttons, a 95+ dB alarm, easy bump-check mode with 24-hour checkmark indicator and large high contrast LCD for easy operation in any work environment, even low-light conditions. Durable, rubberized

housing allows secure grip, can withstand a 20-ft (6.1 m) drop and is both dust- and water-tight.

Uses industry leading MSA XCell™ Sensors for LEL, O₂, two-toxic H₂S/CO, or individual H₂S and CO sensors. The ALT AIR® 4X Multigas Detector is compatible with the ALT AIR® 4 QuickCheck® Station and the ALT AIR® 4 GALAXY® Test Stand (GALAXY® System software upgrade is required).

UL and CSA approved for Class I, Division 1, Groups A, B, C and D; Class II, Division 1, Groups E, F and G; Class III, Division 1, Tamb = -40°C to +54°C; T4

ALT AIR 4X DETECTOR with 3-year warranty, data logging, charger, calibration cap and tubing

Cat. No.	Configuration	Case Color	Rental	Purchase Price
MSA10107602	LEL, O ₂ , CO, H ₂ S	Charcoal	YES	\$945.00
MSA10107603	LEL, O ₂ , CO, H ₂ S	Phosphorescent	YES	966.00
MSA10110443	LEL, O ₂ , CO	Charcoal	YES	899.00
MSA10110445	LEL, O ₂ , CO	Phosphorescent	YES	919.00
MSA10110444	LEL, O ₂	Charcoal	YES	729.00
MSA10110446	LEL, O ₂	Phosphorescent	YES	749.00

ALT AIR 4X FOUR-GAS DETECTOR in black case with calibration cylinder and 0.25 lpm regulator, charger, calibration cap and tubing

Cat. No.	Configuration	Case Color	Rental	Purchase Price
MSA10110488	LEL, O ₂ , CO, H ₂ S	Charcoal	YES	\$1,501.00
MSA10110489	LEL, O ₂ , CO, H ₂ S with Universal Pump Probe	Charcoal	YES	1,948.00

CALIBRATION GAS

Cat. No.	Configuration	Purchase Price
MSA10048280	34 L, 1.45% CH ₄ , 15% O ₂ , 60 ppm CO, 20 ppm H ₂ S	\$254.00
MSA10045035	54 L, 1.45% CH ₄ , 15% O ₂ , 60 ppm CO, 20 ppm H ₂ S	379.00

REPLACEMENT SENSORS

Cat. No.	Description	Purchase Price
MSA10106722	XCell Ex Combustible Sensor	\$223.00
MSA10106729	XCell O ₂ Sensor	199.00
MSA10106725	XCell CO/H ₂ S Two-Tox Sensor	271.00

ACCESSORIES

Cat. No.	Description	Purchase Price
MSA10055576	Universal Pump Probe, CSA	\$465.00
MSA10046528	Universal Pump Probe, North America	465.00



Orion® Multigas Detector

Simultaneously detects the presence of O₂, H₂S, CO and combustible gases. Monitor features intuitive one-button, automatic calibration and a large, backlit display. Housing is IP54-rated for water resistance and dust penetration. Galaxy® System (see page 112) and Link™ software compatible. Datalogging optional.

Designed to meet intrinsic safety testing requirements in certain hazardous atmospheres.

Approved by UL and cUL for use in Class I, Division I, Group A, B, C and D.

Economy Kits include detector, NiMH or alkaline battery pack, charger and calibration cap.

Standard Kits include detector, NiMH battery or alkaline battery pack, internal pump, 10-ft (3.1 m) line and 1-ft (0.30 m) probe.

Deluxe Kits include detector, NiMH battery or alkaline battery pack, internal pump, 10-ft (3.1 m) line, 1-ft (0.30 m) probe, calibration case and a black case.



ECONOMY KITS

Cat. No.	Description	Rental
MSA10026501	LEL, O ₂ , CO, Alkaline	YES
MSA10026500	LEL, O ₂ , H ₂ S, Alkaline	YES
MSA10026499	LEL, O ₂ , CO, H ₂ S, Alkaline	YES
MSA10030395	LEL, O ₂ , CO, H ₂ S, NiMH	YES

STANDARD KITS

Cat. No.	Description	Rental
MSA10026497	LEL, O ₂ , CO, NiMH	YES
MSA10026495	LEL, O ₂ , H ₂ S, NiMH	YES
MSA10026493	LEL, O ₂ , CO, H ₂ S, NiMH	YES
MSA10030421	LEL, O ₂ , CO, H ₂ S, Less Case	YES
MSA10026498	LEL, O ₂ , CO, Alkaline	YES
MSA10026496	LEL, O ₂ , H ₂ S, Alkaline	YES
MSA10026494	LEL, O ₂ , CO, H ₂ S, Alkaline	YES

DELUXE KITS

Cat. No.	Description	Rental
MSA10026491	LEL, O ₂ , CO, NiMH	YES
MSA10026369	LEL, O ₂ , H ₂ S, NiMH	YES
MSA10026367	LEL, O ₂ , CO, H ₂ S, NiMH	YES
MSA10030394	LEL, O ₂ , CO, H ₂ S, Datalog	YES
MSA10026492	LEL, O ₂ , CO, Alkaline	YES
MSA10026490	LEL, O ₂ , H ₂ S, Alkaline	YES
MSA10026368	LEL, O ₂ , CO, H ₂ S, Alkaline	YES



Sirius® PID Multigas Monitor

Simultaneously monitors volatile organic compounds with low vapor pressures and combustible, toxic and oxygen-deficient atmospheres. Monitor can be used with or without 10.6 eV PID mode for built-in confined space monitoring capability.

Monitor features rechargeable lithium-ion battery pack with vehicle charger or replaceable alkaline battery options, one-button, automatic calibration and visual and audible alarms.

Economy Kits include monitor, standard alkaline battery pack (lithium-ion optional) and a standard cap.

Industrial Kits include monitor, standard lithium-ion battery, 10-ft (3.1 m) line, 1-ft (0.30 m) sample probe, black boot, cordura jacket and standard cap. Datalogging and optional calibration kit for 4-gas version only.

Deluxe Kits include monitor, standard lithium-ion battery, 10-ft (3.1 m) line, 1-ft (0.30 m) sample probe, calibration kit, black boot, cordura jacket, standard cap and a black case. Datalogging optional for 4-gas version only.



ECONOMY KITS

Cat. No.	Description	Rental	Purchase Price
MSA10051153	LEL, O ₂ , CO, PID (VOCs), Alkaline	YES	\$3,518.00
MSA10051152	LEL, O ₂ , H ₂ S, PID (VOCs), Alkaline	YES	3,518.00
MSA10051151	LEL, O ₂ , CO, H ₂ S, PID (VOCs), Alkaline	YES	3,723.00
MSA10051154	LEL, O ₂ , CO, H ₂ S, PID (VOCs), Li-ion	YES	3,798.00

INDUSTRIAL KITS

Cat. No.	Description	Rental	Purchase Price
MSA10051148	LEL, O ₂ , CO, PID (VOCs)	YES	\$3,974.00
MSA10051147	LEL, O ₂ , H ₂ S, PID (VOCs)	YES	3,974.00
MSA10051146	LEL, O ₂ , CO, H ₂ S, PID (VOCs)	YES	4,180.00
MSA10051150	LEL, O ₂ , CO, H ₂ S, PID (VOCs) with Datalogging and Calibration Kit	YES	5,185.00

DELUXE KITS

Cat. No.	Description	Rental	Purchase Price
MSA10051143	LEL, O ₂ , CO, PID (VOCs)	YES	\$4,732.00
MSA10051142	LEL, O ₂ , H ₂ S, PID (VOCs)	YES	4,732.00
MSA10051141	LEL, O ₂ , CO, H ₂ S, PID (VOCs)	YES	4,938.00
MSA10051144	LEL, O ₂ , CO, H ₂ S, PID (VOCs) with Datalogging	YES	5,317.00



GasAlertQuattro Multi-Gas Detector

• Visual auditing, easy compliance

Displays real-time gas concentrations readings for up to four gas hazards including H₂S, CO, O₂ and combustible gases (%LEL). Audible, visual and vibrating alarms alert users to unsafe conditions while downloadable datalogging provides exposure reporting.

Monitor features IntelliFlash, a green LED confidence flash that verifies normal operation to both the user and supervisors from up to 20-ft (6.1 m) and easy operation with one-button simplicity. Monitors are available in alkaline (3 AA) or rechargeable battery models. Additional features include multi-language support in English, French, German, Spanish and Portuguese and easily

identifiable bump test and calibration icons. Compatible with MicroDock II (see page 114) for easy calibration, test and record storage.

4-GAS DETECTORS

Cat. No.	Description	Rental	Purchase Price
BWTQTXWHMAYNA	O ₂ , LEL, H ₂ S, CO; Alkaline Battery	YES	\$695.00
BWTQTXWHMRYNA	O ₂ , LEL, H ₂ S, CO; Rechargeable Battery	YES	771.00

3-GAS DETECTORS

Cat. No.	Description	Rental	Purchase Price
BWTQTXWHOAYNA	O ₂ , LEL, H ₂ S; Alkaline	YES	\$645.00
BWTQTXWHOYNA	O ₂ , LEL, H ₂ S; Rechargeable	YES	705.00
BWTQTXWOMAYNA	O ₂ , LEL, CO; Alkaline	YES	645.00
BWTQTXWOMRYNA	O ₂ , LEL, CO; Rechargeable	YES	705.00
BWTQTXOHMAYNA	O ₂ , H ₂ S, CO; Alkaline	YES	645.00
BWTQTXOHMRYNA	O ₂ , H ₂ S, CO; Rechargeable	YES	705.00
BWTQTOWHMAYNA	LEL, H ₂ S, CO; Alkaline	YES	645.00
BWTQTOWHMRYNA	LEL, H ₂ S, CO; Rechargeable	YES	705.00

2-GAS DETECTORS

Cat. No.	Description	Rental	Purchase Price
BWTQTXW00AYNA	O ₂ , LEL; Alkaline	YES	\$595.00
BWTQTXW00RYNA	O ₂ , LEL; Rechargeable	YES	671.00
BWTQTO0HMAYNA	H ₂ S, CO; Alkaline	YES	595.00
BWTQTO0HMYNA	H ₂ S, CO; Rechargeable	YES	671.00
BWTQTXO00AYNA	O ₂ , H ₂ S; Alkaline	YES	595.00
BWTQTXO00RYNA	O ₂ , H ₂ S; Rechargeable	YES	671.00
BWTQTXO0MAYNA	O ₂ , CO; Alkaline	YES	595.00
BWTQTXO0MRYNA	O ₂ , CO; Rechargeable	YES	671.00
BWTQTOWH0AYNA	LEL, H ₂ S; Alkaline	YES	595.00
BWTQTOWH0RYNA	LEL, H ₂ S; Rechargeable	YES	671.00
BWTQTOW0MAYNA	LEL, CO; Alkaline	YES	595.00
BWTQTOW0MRYNA	LEL, CO; Rechargeable	YES	671.00

SINGLE GAS DETECTORS

Cat. No.	Description	Rental	Purchase Price
BWTQTX000AYNA	O ₂ ; Alkaline	YES	\$545.00
BWTQTX000RYNA	O ₂ ; Rechargeable	YES	605.00
BWTQTO000AYNA	LEL; Alkaline	YES	545.00
BWTQTO000RYNA	LEL; Rechargeable	YES	605.00
BWTQTO00H0AYNA	H ₂ S; Alkaline	YES	545.00
BWTQTO00H0RYNA	H ₂ S; Rechargeable	YES	605.00
BWTQTO00MAYNA	CO; Alkaline	YES	545.00
BWTQTO00MRYNA	CO; Rechargeable	YES	605.00



GasAlertMax XT II Portable Gas Detector



Features an integrated sampling pump designed with SmartSample technology to prevent moisture from entering pump and to save time with reliable pump block detection. Capable of monitoring up to four hazards including H₂S, CO, O₂ and combustibles. Offers easy operation with one-button simplicity. Compact, comfortable and field tough with built-in rechargeable lithium battery and built-in concussion-proof boot. Best in class block detection with semi-conductor pressure sensor. Detector can be conveniently managed using Fleet Manager II software, and is equipped with standard datalogging and event logging. Additional features include multi-language support in English, French, German, Spanish and Portuguese; low, high STEL (short-term exposure limit), TWA (time-weighted average) and OL (over limit) alarms; and visual, audible and

vibrating alarms. Compatible with MicroDock II (see page 114) for easy calibration, test and record storage.

Standard accessories include stainless-steel alligator clip, rechargeable battery, charging adaptor, 3-ft (0.91 m) calibration hose with quick-connect, 10-ft (3.1 m) sampling hose with particulate filter and quick-connect, reversible screwdriver, pump filter spares and instructions and laminated quick reference card.

Deluxe Confined Space Kit includes concussion-proof boot, carrying holster, IR connectivity kit, 0.5 LPM regulator, 3-ft (0.91 m) calibration hose with quick-connect, 10-ft (3.1 m) sampling hose with quick-connect, sampling probe and hard-sided carrying case with lid insert pockets and foam case insert.

YELLOW DETECTOR – Also available in Black

Cat. No.	Description	Rental	Purchase Price
BWTXTXWHMRYNA	LEL, O ₂ , H ₂ S, CO	YES	\$865.00
BWTXTXWHOYNA	LEL, O ₂ , H ₂ S	YES	815.00
BWTXTXWOMRYNA	LEL, O ₂ , CO	YES	815.00
BWTXTXOHMRYNA	O ₂ , H ₂ S, CO	YES	815.00
BWTXTXWHMRYNA	LEL, H ₂ S, CO	YES	815.00
BWTXTXW00RYNA	LEL, O ₂	YES	771.00
BWTXTXO0HMYNA	H ₂ S, CO	YES	771.00
BWTXTXO0H0RYNA	O ₂ , H ₂ S	YES	771.00
BWTXTXO0MRYNA	O ₂ , CO	YES	771.00
BWTXTXO0H0RYNA	LEL, H ₂ S	YES	771.00
BWTXTXO0MRYNA	LEL, CO	YES	771.00
BWTXTXO00RYNA	O ₂	YES	695.00
BWTXTXO00RYNA	LEL	YES	695.00
BWTXTXO0H0RYNA	H ₂ S	YES	695.00
BWTXTXO00MYNA	CO	YES	695.00

ACCESSORIES

Cat. No.	Description	Rental	Purchase Price
BWTXTXWHMRYNACS	Deluxe Confined Space Kit	YES	\$1,295.00



GasAlertMicro5 Portable Gas Detector

Simultaneously monitors and displays up to 5 potential atmospheric hazards including O₂, LEL and a range of toxic gases. EMI and RFI resistant housing is IP65/67-rated for water-resistance and its highly concussion-proof enclosure. Audible, visual and vibrating alarms activate in the event of a low, high, TWA or STEL alarm conditions. Features passcode protection, backlight LCD display, combustible correction factors, multi-language support and automatic calibration.

Comes standard with alkaline batteries or a rechargeable battery pack. Compatible with MicroDock II testing and calibration station (see page 114).

Standard Kits include calibration adaptor and hose, quick reference guide and technical documentation CD, integral concussion-proof boot and alkaline batteries.

Deluxe Confined Space Kits include carrying holster, remote sampling kit with 10-ft (3.1 m) sample tubing and probe, manual aspirator pump, IR connectivity kit, Fleet Manager II CD, regulator, fittings, auxiliary filters and foam-lined carrying case.

Diffused models also available; contact Total Safety for ordering information.

2-year warranty including sensors.

Class I, Division 1, Groups A, B, C and D.

STANDARD KIT, PUMPED

Cat. No.	Description	Rental	Purchase Price
BWTM5XW0YAPDYN00	LEL, O ₂ , CO, H ₂ S	YES	\$1,130.00
BWTM5XWSYAPDYN00	LEL, O ₂ , CO, H ₂ S, SO ₂	YES	1,425.00
BWTM5XWCYAPDYN00	LEL, O ₂ , CO, H ₂ S, Cl ₂	YES	1,425.00

WITH PID, PUMPED

Cat. No.	Description	Rental	Purchase Price
BWTM5PID00Q0APDYN00	VOCs	YES	\$1,220.00
BWTM5PIDXWQCAPDYN00	VOCs, LEL, O ₂ , Cl ₂	YES	1,790.00
BWTM5PIDXWQSAPDYN00	VOCs, LEL, O ₂ , SO ₂	YES	1,790.00
BWTM5PIDXWQAAPDYN00	VOCs, LEL, O ₂ , NH ₃	YES	1,890.00
BWTM5PIDXWQYAPDYN00	VOCs, LEL, O ₂ , CO, H ₂ S	YES	1,745.00

WITH IR, PUMPED

Cat. No.	Description	Purchase Price
BWTM5IR00B0APDYN00	CO ₂	\$1,430.00
BWTM5IRXWB0APDYN00	CO ₂ , LEL, O ₂	1,705.00
BWTM5IRXWBHAPDYN00	CO ₂ , LEL, O ₂ , H ₂ S	1,895.00
BWTM5IRXWBYPDYN00	CO ₂ , LEL, O ₂ , CO, H ₂ S	1,955.00

ACCESSORIES

Cat. No.	Description	Purchase Price
BWTM5CKDL	Deluxe Confined Space Kit	\$325.00
BWTM5CKCC	Carrying Case with Foam	145.00



GasAlertMicroClip XT Portable Gas Detector

Features visual compliance with IntelliFlash™ and simple one-button operation, which offers ease of use and significantly reduces time spent training the user. Capable of monitoring up to 4 hazards including H₂S, CO, O₂ and combustibles.

Rugged and durable with built-in concussion-proof boot, yet thin and lightweight for wearer comfort. Detector can be conveniently managed using Fleet Manager II software, and powered by a lithium polymer battery. Additional features include multi-language support in English, French, German, Spanish and Portuguese; low, high STEL (short-term exposure limit), TWA (time-weighted average) and OL (over limit) alarms; and visual, audible and vibrating alarms. Compatible with MicroDock II (see page 114) for easy calibration, test and record storage.

Deluxe Confined Space Kit includes concussion-proof boot, carrying holster, IR connectivity kit, 0.5 LPM regulator, 3-ft (0.91 m) calibration hose with quick-connect, 10-ft (3.1 m) sampling hose with quick-connect, sampling probe and hard-sided carrying case with lid insert pockets and foam case insert.



YELLOW DETECTOR – Also available in Black

Cat. No.	Description	Rental	Purchase Price
BWTMC2XWHMYNA	O ₂ , LEL, H ₂ S, CO	YES	\$595.00
BWTMC2XWH0YNA	O ₂ , LEL, H ₂ S	YES	520.00
BWTMC2XW0MYNA	O ₂ , LEL, CO	YES	520.00
BWTMC2X0HMYNA	O ₂ , H ₂ S, CO	YES	520.00
BWTMC20WHMYNA	LEL, H ₂ S, CO	YES	520.00
BWTMC2XW00YNA	O ₂ , LEL	YES	435.00
BWTMC20WH0YNA	LEL, H ₂ S	YES	435.00
BWTMC200HMYNA	H ₂ S, CO	YES	435.00
BWTMC2X0H0YNA	O ₂ , H ₂ S	YES	435.00
BWTMC2X00MYNA	O ₂ , CO	YES	435.00
BWTMC20W0MYNA	LEL, CO	YES	435.00
BWTMC2X000YNA	O ₂	YES	360.00
BWTMC20W00YNA	LEL	YES	385.00
BWTMC200H0YNA	H ₂ S	YES	385.00
BWTMC2000MYNA	CO	YES	389.00



QRAE II Multigas Detector



A full-featured, compact, 1- to 4-sensor, diffusion or pumped, gas detector for H₂S, CO, O₂ and combustibles. Highly rated for EMI/RFI immunity technology for eliminating radio interference. Patented SPE O₂ sensor provides extra life performance with a RoHS compliant leadfree design. Lithium-ion battery model delivers up to 14 hours of continuous operation in diffusion mode (a spare alkaline batter adapter is included for 10-hour back-up operation).

Detectors include datalogging storage capacity for download to PC-compatible computers, a backlit LCD display, easy-to-change

external battery packs, easy access sensors, internal vibrating and audible alarm and IP65 water- and dust-resistant case. AutoRAE Lite compatible (see page 116).

2-year warranty. UL approved. CSA certified.

Cat. No.	Description	Rental	Purchase Price
RAE02011111A0	Diffusion LEL, O ₂ , H ₂ S, CO, Alkaline	YES	\$563.00
RAE02011110A0	Diffusion LEL, O ₂ , H ₂ S, CO, Li-ion	YES	615.00
RAE02011112A0	Pumped LEL, O ₂ , H ₂ S, CO, Li-ion	YES	718.00
RAE02011111AA	10-Unit Pack – Diffusion LEL, O ₂ , H ₂ S, CO, Alkaline	YES	4,630.00
RAE02011110AA	10-Unit Pack – Diffusion LEL, O ₂ , H ₂ S, CO, Li-ion	YES	5,145.00
RAE02011112AA	10-Unit Pack – Pumped LEL, O ₂ , H ₂ S, CO, Li-ion	YES	6,587.00
RAE0203461005	5-Unit Charger Bump/Calibration Station		407.00

ACCESSORIES

Cat. No.	Description	Purchase Price
RAET010001000	AutoRAE Lite for QRAE II	\$1,334.00



MultiRAE Pro Monitor



Combines continuous monitoring capabilities for gamma radiation, volatile organic compounds (VOCs), oxygen, toxic and combustible gasses.

Features a large graphical display with easy-to-use, icon-driven user interface. Intelligent sensors store calibration data, so they can be swapped in the field. Wireless capability elevates worker protection to the next level by providing safety officers real-time access to instrument readings and alarm status from any location for better visibility and faster incident response. Other features include fully automated bump testing and calibration with AutoRAE 2 (contact Total Safety for ordering information) and unmistakable

5-way local and remote wireless notification of alarm conditions, including the Man Down Alarm, which sounds when the motion sensor inside the monitor registers no movement in a certain amount of time. The wireless and consecutive alarm mode in the remote command center or safety headquarters allows more people to know and respond in the event of an emergency creating a safer worksite. Monitor also features over 30 interchangeable sensor options, including parts-per-billion PID.

Includes monitor with sensors, pump, battery and non-wireless options and protective rubber boot, swivel belt clip and filter installed. Wireless option available. Contact Total Safety for more information. Continuous datalogging (6 months for 5 sensors at 1-min intervals), travel charger/PC communication cable, AC adapter, calibration adapter, alkaline battery adapter, hex tool, PID sensor cap removal tool, PID zeroing filter, IP-67/decontamination valve accessory, three spare external filters, 6-in (15.2 cm) inlet probe, quick start guide, CD with documentation, ProRAE Studio II Instrument Configuration and Data Management Software, calibration certificate and test certificate.

2-year warranty. UL approved. CSA Certified.

PUMPED /10.6 EV PPB PID/LEL/CO + H₂S/O₂/GAMMA/LI-ION

Cat. No.	Description	Purchase Price
RAEMCB3A3C1REZ020	Unit only	\$6,193.00
RAEMCB3A3C1REZ02M	Unit with Accessories/Confined Space and Calibration Kit (4-gas + 10 ppm Iso)	7,003.00
RAEMCB3A3C1REZ02N	Unit with Accessories/Confined Space and Calibration Kit (4-gas + 10 ppm Iso + 100 ppm Iso)	7,236.00

PUMPED/10.6 EV PPB PID/LEL/CL/NH /O₂/LI-ION

Cat. No.	Description	Purchase Price
RAEMCB3A3C168E020	Unit only	\$5,493.00





**INDUSTRIAL
SCIENTIFIC**

Ventis™ MX4 Multigas Monitor

Compact, lightweight detector provides reliable monitoring for O₂, LEL, CO and H₂S gases with a wide range of sensor options. May be configured for diffusion applications or with an integral sampling pump for sample draw applications. Features LCD readout, peak-hold function, STEL, TWA and datalogging. Diffusion monitor features a rechargeable lithium-ion extended range battery pack with a 20 hour run-time. Constructed of high-impact polycarbonate housing and rubber overmold that help absorb impact from accidental drops. Extremely water-resistant—third party certified IP66 and IP67.



Confined Space Kits include choice of Aspirated Ventis MX4 monitor, universal charger, soft carrying case, service tool, reference guide, calibration tubing, dust filter/water stop, calibration fitting, sample tubing, calibration gas (appropriate mix) with regulator and rugged Pelican® case. Comes standard in black.

UL approved and CSA certified for Class I, Division 1, Groups A, B, C, D, Class II, Groups F, G

COMMON INSTRUMENT CONFIGURATIONS*

Cat. No.	Description	Rental	Purchase Price
ISCVTSL1231100101	LEL (Methane), CO, H ₂ S, O ₂ , Li-Ion, Desktop Charger	YES	\$695.00
ISCVTSK5231300111	LEL (Pentane), SO ₂ , H ₂ S, O ₂ , Li-Ion, Auto Charger, Soft Case	YES	834.00
ISCVTSL1232110121	LEL (Pentane), CO, NO ₂ , O ₂ , Li-Ion, No Charger, Soft Case	YES	960.00
ISCVTSK1232110111	With Pump: LEL (Pentane), CO, H ₂ S, O ₂ , Ext. Li-Ion, Desktop Charger, Soft Case	YES	910.00
ISCVTSL1232100121	With Pump: LEL (Methane), CO, H ₂ S, O ₂ , Ext. Li-Ion, Desktop Charger, Hard Case w/Display Window	YES	760.00

*Custom configurations also available, please contact Total Safety for more information.

CONFINED SPACE KITS

Cat. No.	Description	Rental	Purchase Price
ISCVKK123211XY1Z	LEL, CO, H ₂ S, O ₂	YES	Call for Quote
ISCVKK103211XY1Z	LEL, CO, O ₂	YES	Call for Quote
ISCVKK023211XY1Z	LEL, H ₂ S, O ₂	YES	Call for Quote
ISCVKK003211XY1Z	LEL, O ₂	YES	Call for Quote

X = color 1 = Black 0 = Orange; Y = Certification 1 = UL/CSA; Z = Language 1 = English

OPTIONAL ACCESSORIES

Cat. No.	Description	Rental	Purchase Price
ISC181086300BC	DS2 Docking Station for Ventis MX4	YES	Call for Quote
ISC1810863100	V-Cal Calibration Station, No Pump	YES	874.50
ISC18107664ABC	V-Cal 6 Unit Calibration Station	YES	Call for Quote
ISC18108191	Single-Unit Charger	YES	70.00
ISC181086500	6-Unit Charger	YES	445.00

**INDUSTRIAL
SCIENTIFIC**

iNet® Gas Detection Service



iNet keeps people safe by providing visibility into alarms, exposure and usage. It keeps gas detectors working without costly and time-consuming maintenance. And there is no need to purchase gas detectors when you subscribe to iNet; gas detection is received as a service.

iNet Control Software

The first hosted software for managing gas detector fleets and included with every iNet subscription. It provides visibility into alarms, exposure and usage and allows users to view trend graphs or sensor-level detail for each gas detector. iNet Control also allows users to compare the health of their gas detection program to industry averages.

iNet DS

The iNet DS docking station connects directly to the Internet via an Ethernet interface and does not require any installation of software or server hardware at the customer site. It offers seamless gas detector fleet configuration and manageability using iNet Control on any PC Web browser. Instrument firmware upgrades are done automatically, and the system allows mobile operation.

iNet InSite

iNet InSite offers a plug-and-play docking station solution for companies who own their gas detectors and have the staff needed to maintain and service them. With a fixed monthly subscription to iNet InSite (which includes iNet DS docking stations), users can configure and manage their gas detection fleet with unlimited access to iNet Control.

iNet customers reduce their instrument fleet by an average of ~20% because of limited downtime.

Contact Total Safety for more information about iNet Gas Detection Services.

***All ABC extensions require:**

+ DS2 Docking Station Ordering Information

A = Wireless Option (currently unavailable) 0 – none

B = Number of iGas Readers

C = Power Cord Option

0 – US 1 – UK 2 – EU 3 – AUS
4 – ITA 5 – DEN 6 – SWZ


**INDUSTRIAL
SCIENTIFIC**
MX6 iBrid™ Multigas Monitor


Detects 1-to-6 gases, including options for PID (VOCs) and 24 available sensors including sensors for combustible gases and CO₂. Full-color LCD display screen allows for easy viewing in low-to-bright light conditions. Intuitive menu guides the user through instrument settings and functions with 5-way navigation button. Supports the option of on-board graphing for easily interpreted direct readings and recorded data.

Configured either with an integral sampling pump or for diffusion operation, unit is ideal for applications from confined space entry to personal survey monitoring. Fully compatible with our DS2 Docking Station™ (see page 114)

and iNet™ Instrument Network for remote area monitoring. Supports up to 36 hours with rechargeable extended runtime (ER) battery; however, other battery options are available.

Confined Space Kits include monitor, rechargeable lithium-ion battery pack and additional accessories. Contact Total Safety for a complete list of accessories.

Lifetime warranty. UL approved and CSA certified Class I, Groups A, B, C and D; T4.

Cat. No.	Description	Rental	Purchase Price
ISCMX6K0030101	LEL, O ₂ , Li-ion, Diff	YES	\$1,550.00
ISCMX6K1030101	LEL, CO, O ₂ , Li-ion, Diff	YES	1,860.00
ISCMX6K0230101	LEL, H ₂ S, O ₂ , Li-ion, Diff	YES	1,860.00
ISCMX6K1230101	LEL, CO, H ₂ S, O ₂ , Li-ion, Diff	YES	2,170.00
ISCMX6K0030301	LEL, O ₂ , Alk, Diff	YES	1,465.00
ISCMX6K1030301	LEL, CO, O ₂ , Alk, Diff	YES	1,775.00
ISCMX6K0230301	LEL, H ₂ S, O ₂ , Alk, Diff	YES	1,775.00
ISCMX6K1230301	LEL, CO, H ₂ S, O ₂ , Alk, Diff	YES	2,085.00
ISCMX6K1230211	LEL, CO, H ₂ S, O ₂ , Li-ion/ER, w/Pump		2,530.00
ISCMX6K103Q211	LEL, CO, O ₂ , CO ₂ , Li-ion/ER, w/Pump		2,740.00
ISCMX6K123R211	LEL, CO, H ₂ S, O ₂ , PID, Li-ion/ER, w/Pump		3,090.00

CONFINED SPACE KITS

Cat. No.	Description	Rental	Purchase Price
ISCMX6KITK1230211	LEL, CO, H ₂ S, O ₂ w/Pump	YES	\$3,400.00
ISCMX6KITK123R211	LEL, CO, H ₂ S, O ₂ , PID w/Pump	YES	4,110.00

ACCESSORIES

Cat. No.	Description	Rental	Purchase Price
ISC18107136	Battery Charger, 5-Unit		\$460.00
ISC18107094	Battery Charger/Datalink	YES	390.00
ISC18107011	Battery Charger, 12V		145.00
ISC18106765	SP6 Motorized Sampling Pump Module	YES	330.00
ISC18106724ABC	DS2 Instrument Docking Station for MX6	YES	2,210.00

**INDUSTRIAL
SCIENTIFIC**
**BM25 Multiguard Transportable
Multigas Area Monitor**


Capable of simultaneously monitoring from 1-to-5 gases through diffusion or sample draw. The impact resistant monitor is designed for team protection or area surveillance, perimeter monitoring, rig overhauls, and mobile or short-term work in areas where fixed detection systems are not suitable.

5-gas configuration features STEL and TWA readings and nearly 4 months of datalogging capacity. Long-term area monitoring is possible in hazardous locations with an intrinsically safe trickle charge power supply. Capable of wired interconnection with other BM25 units. Comes with a backlit LCD display, a NiMH battery pack, a 120V universal charger and power cord, dry contact alarm and fault outputs and audible alarm.

CSA certified for Class I, Groups A, B, C and D; T4.

WITH DIFFUSION DRAW

Cat. No.	Description	Purchase Price
ISC6514872K00300	LEL, O ₂	\$3,495.00
ISC6514872K02300	LEL, H ₂ S, O ₂	3,805.00
ISC6514872K10300	LEL, CO, O ₂	3,805.00
ISC6514872K12300	LEL, CO, H ₂ S, O ₂	4,115.00
ISC6514872K03J50	LEL, O ₂ , CO, H ₂ S, SO ₂	4,265.00
ISC6514872K02350	LEL, H ₂ S, O ₂ , SO ₂	4,175.00
ISC6514872K103Q0	LEL, CO, O ₂ , CO ₂	4,520.00
ISC6514872K67300	LEL, NH ₃ , Cl ₂ , O ₂	4,325.00
ISC6514872K09J30	LEL, PH ₃ , CO, H ₂ S, O ₂	4,555.00

WITH PUMP DRAW

Cat. No.	Description	Purchase Price
ISC6514872K02301	LEL, H ₂ S, CO	\$4,480.00
ISC6514872K12301	LEL, CO, H ₂ S, O ₂	4,790.00

ACCESSORIES

Cat. No.	Description	Purchase Price
ISC6111303	Intrinsically Safe Power Supply	\$329.00
ISCWLOGUSB	Datalink Adapter Kit (Software with USB Adapter Cable)	763.00

Honeywell MultiPro™ Multigas Detector

Offers real-time readings of O₂, combustibles (LELs) and a choice of CO, H₂S or up to 2 toxic channels using Duo-Tox two channel CO/H₂S sensor. Incorporates one-button operation (including calibration) and a built-in event logger.

Features a backlit display, IrDA port for communication with PC, IPX5-rated for protection against water ingress and exceeds IEC 1000-4-3 for level 3 RFI immunity specifications. Includes STEL and TWA visible and audible alarms. Vibrating alarm optional. Black-box recorder standard. Datalogging optional. IQ Express compatible (see page 117).

Standard accessories include rechargeable lithium-ion battery with charger or alkaline batteries, clothes clip, laminated quick reference guide and calibration adapter.

UL approved for Class I, Division 1, Groups A, B, C and D; T4; Class II, Division 1, Groups E, F and G; Class III. CSA certified for Class I, Division 1, Groups A, B, C and D; T4.

For a list of all compliances, please contact Total Safety.



Cat. No.	Description	Rental	Purchase Price
SPR5448300A	LEL, O ₂ , Alkaline	YES	\$467.00
SPR5448301A	LEL, O ₂ , CO, Alkaline	YES	589.00
SPR5448302A	LEL, O ₂ , H ₂ S, Alkaline	YES	589.00
SPR5448314A	LEL, O ₂ , Duo-Tox, Alkaline	YES	600.00
SPR5448300N	LEL, O ₂ , Li-ion	YES	590.00
SPR5448301N	LEL, O ₂ , CO, Li-ion	YES	700.00
SPR5448302N	LEL, O ₂ , H ₂ S, Li-ion	YES	700.00
SPR5448314N	LEL, O ₂ , Duo-Tox, Li-ion	YES	730.00



...to ensure the safe
Wellbeing of Workers Worldwide



EAGLE™ Gas Detector



Simultaneously monitors up to 6 gases including a maximum of 2 super toxic gases. Monitor features over 350 monitoring configurations, an automatically backlit LCD display, PPM or LEL hydrocarbon detection, automatic calibration, RFI and EMI, chemical- and weather-resistant housing. Features 2 alarms per channel plus TWA and STEL alarms for toxics. Visual and audible alarms are fully adjustable for levels, latching or self reset and silenceable.

Monitor comes standard with alkaline batteries (rechargeable NiCad batteries optional), a hydrophobic probe and 5-ft (1.5 m) hose. An internal hydrophobic filter is included in most versions. Certain toxic versions are equipped with a special probe, inlet fitting and 3-ft (0.91 m) Teflon® hose. For HF and O₂ versions, 3-ft (0.91 m) Teflon hose is to be used without probe. Datalogging of up to 4 gases optional.

Intrinsically safe, Class I, Division 1, Groups A, B, C and D. Most versions CSA and UL approved.

Standard Confined Space Gases: CH₄, O₂, CO and H₂S.

Super Toxics and Other Gases: NH₃, AsH₃, CO₂, Cl₂, ClO₂, F₂, HF, HCl, HCN, H₂S, CH₄, IR, C₄H₁₀, IR, NO₂, NO, O₃, PH₃, SiH₄ and SO₂.

Cat. No.	Description	Rental	Purchase Price
RKI725401RK	LEL/PPM, O ₂ , H ₂ S, CO	YES	\$2,250.00
RKI725402RK	LEL/PPM, O ₂ , H ₂ S, SO ₂		2,900.00
RKI725403RK	LEL/PPM, O ₂ , CO, SO ₂		2,900.00
RKI725404RK	LEL/PPM, O ₂ , H ₂ S, Cl ₂		2,900.00
RKI725442RK	LEL/PPM, CH ₄ , IR, O ₂ , H ₂ S		2,900.00
RKI725451RK	LEL, CH ₄ , IR, O ₂ , CO, H ₂ S		2,900.00
RKI725335RK05	LEL/% Vol, CH ₄ , O ₂ , CO ₂		3,350.00
RKI725455RK02	LEL/% Vol, CH ₄ , O ₂ , H ₂ S, CO ₂		2,900.00

ACCESSORIES

Cat. No.	Description	Purchase Price
RKI570012RK	Datalogging Board	\$140.00
RKIRK1037	NiCad Battery, 4 Required	130.00
RKI492150RK	Battery Charger, 115V	115.00


MSA
ALTAIR® Pro Single Gas Detector


Single gas monitor capable of detecting CO, H₂S, O₂, NH₃, Cl₂, ClO₂, NO₂, SO₂, HCN or PH₃. Features rubberized housing that is rated IP67 against water and dust ingress (except O₂-R, which is IP54-rated), large backlit LCD display, standard event and datalogging capabilities and adjustable alarm set points are offered for low, high, TWA and STEL indicated by visual, audible and vibrating alarms. Replaceable CR2 camera battery and sensor. Galaxy® Automated Test System (see page 112), ALTAIR® QuickCheck™ Station (see page 112) and FiveStar® Link® Software compatible.

Features a remote oxygen detector (O₂-R) version that is attached remotely at the end of a 10-ft (3.1 m) cable, rather than inside the instrument. This design allows the user to lower the sensor into a remote location and monitor the oxygen levels from 10-ft (3.1 m) away.

UL approved and CSA certified for Class I, Division 1, Groups A, B, C and D.

Cat. No.	Description	Rental	Purchase Price
MSA10074137	O ₂		\$421.00
MSA10074135	CO	YES	421.00
MSA10076723	CO Fire		421.00
MSA10076724	CO Steel		535.00
MSA10074136	H ₂ S	YES	421.00
MSA10076729	HCN	YES	546.00
MSA10076716	Cl ₂	YES	546.00
MSA10076717	ClO ₂	YES	667.00
MSA10076736	SO ₂	YES	541.00
MSA10076731	NO ₂		667.00
MSA10076730	NH ₃	YES	667.00
MSA10076735	PH ₃		667.00
MSA10076733	O ₂ -R		667.00

ACCESSORIES

Cat. No.	Description	Purchase Price
MSA467895	Regulator, 0.25 LPM	\$207.00
MSA602294	Tubing, 16-in (Not for Use with NH ₃ , Cl ₂ and ClO ₂)	6.00
MSA10080534	Tubing, 16-in Tygon-Lined (for Use with NH ₃ , Cl ₂ and ClO ₂)	27.00
MSA10074132	3V CR2 Battery	10.00

MSA
ALTAIR® Maintenance-Free Single Gas Detector


2-year monitor that features sensor options for CO, H₂S and O₂. Monitor comes standard with rubberized housing that offers IP67 dust and water protection and high resistance to RFI, event recorder that stores up to 25 events automatically and a lithium battery. Features an LCD display, a triple-alarm system with visual, audible and vibrating alarms and built-in calibration and IR communication. Galaxy® Automated Test System (see page 112), ALTAIR® QuickCheck™ Station (see page 112) and MSA Link software compatible.

2-year warranty or 18 hours of alarm.

UL approved and CSA certified for Class I, Division 1, Groups A, B, C and D.

Cat. No.	Description	Rental	Purchase Price
MSA10092522	CO		\$260.00
MSA10092521	H ₂ S	YES	260.00
MSA10092523	O ₂		260.00

ACCESSORIES

Cat. No.	Description	Rental	Purchase Price
MSA710882	Cylinder, 60 ppm CO		\$237.00
MSA473180	Cylinder, 300 ppm CO		207.00
MSA467897	Cylinder, 40 ppm H ₂ S, RP		207.00
MSA711062	Cylinder, 40 ppm H ₂ S, Econo-Cal		171.00
MSA467895	Regulator, 0.25 LPM		207.00
MSA602294	Tubing, 16-in		6.00
MSA10076704	Automatic ALTAIR QuickCheck Station		569.00
MSA10076692	Manual ALTAIR QuickCheck Station		255.00
MSA10078252	Galaxy Automated Test System	YES	1,425.00



GasAlert Extreme Monitor

Designed with durability and comfort in mind, the GasAlert Extreme reliably monitors for any single gas hazard within its wide range of available toxic gas models. With easy on/off operation, this single gas detector offers extended longevity with a 2 year field-replaceable battery and sensor. Calibration is a simple automatic procedure and is compatible with BW Technologies by Honeywell's MicroDock II automatic test and calibration system (see page 114). A wide range of user options, multi-language display and datalogging functionality make the GasAlert Extreme an ideal solution for many applications.

Standard 3V battery provides up to 2 years of continuous operation. Standard accessories include field replaceable battery, stainless-steel alligator clip, test cap, hose and instructions.

Datalogging / Multi-language Versions include all of the above plus full event logging and datalogging capabilities. Downloading datalogged information requires use of the highspeed IR adaptor. Contact Total Safety for pricing.

SamplerPak Kits include standard accessories plus an external motorized sampling pump, sampling hose with airstone filter, coiled connector hose, 2 sampling tubes and a carrying case.

2-year warranty including sensor. 1-year for NH₃ sensor.

UL approved for Class I, Division 1, Groups A, B, C and D.



Cat. No.	Description	Rental	Purchase Price
BWTGAXTHDL	H ₂ S (0-100 ppm)	YES	\$275.00
BWTGAXTH2DL	H ₂ S (0-500 ppm)		310.00
BWTGAXTMDL	CO	YES	275.00
BWTGAXTM2DL	CO (H ₂ Resistant)		410.00
BWTGAXTVDL2	O ₂		275.00
BWTGAXTSDL	SO ₂	YES	410.00
BWTGAXTCDL	Cl ₂	YES	410.00
BWTGAXTZDL	HCN	YES	410.00
BWTGAXTDDL	NO ₂		410.00
BWTGAXTADL	NH ₃ (0-100 ppm)	YES	540.00
BWTGAXTA2DL	NH ₃ (0-400 ppm)		540.00
BWTGAXTPDL	PH ₃		410.00
BWTGAXTGDL	O ₃		735.00
BWTGAXTEDL	ETO		585.00
BWTGAXTNDL	NO		410.00
BWTGAXTVDL	ClO ₂	YES	410.00

ACCESSORIES

Cat. No.	Description	Purchase Price
BWTGASPAK	SamplerPak Motorized Sampling Pump Kit	\$325.00
BWTGASPAKNA	Deluxe SamplerPak Motorized Sampling Pump Kit w/ Rechargeable Batteries & Charger	395.00



GasAlertClip Extreme Monitor (2 & 3 Year Versions)

Now Available:

- User Adjustable Alarm Setpoints
- Low-5ppm/High-10ppm OR
- Low-10ppm/High-15ppm

Simply press the START/TEST button and the GasAlert Extreme is ready to use, providing 2 to 3 years of continuous, maintenance-free operation from the day of activation. Features built-in concussion-proof boot, an LCD display, EMI and RFI resistance, IP66/67-rated housing for water- resistance, automatic event logging records 10 most recent alarm events that can be downloaded via built-in IR communication port to a printer, computer or MicroDock II docking station module (see page 114).

Monitor requires no calibration, sensor replacement, battery replacement or charging. Audible, visual and vibrating alarms come standard. Standard versions include stainless-steel alligator belt clip, test cap and instructions.

Lifetime warranty of instrument including sensors.



2-YEAR VERSIONS

Cat. No.	Description	Rental	Purchase Price
BWTGA24XTH	H ₂ S	YES	\$189.00
BWTGA24XTM	CO	YES	189.00
BWTGA24XTS	SO ₂	YES	310.00
BWTGA24XTX	O ₂	YES	189.00

3-YEAR VERSIONS

Cat. No.	Description	Purchase Price
BWTGA36XTH	H ₂ S	\$240.00
BWTGA36XTM	CO	240.00

ACCESSORIES

Cat. No.	Description	Purchase Price
BWTGAUSB2	IR DataLink USB Adaptor	\$200.00

CALIBRATION EQUIPMENT

Cat. No.	Description	Rental	Purchase Price
BWTDOCK221E00G	MicroDock II Docking Module for H ₂ S	YES	\$1,495.00
BWTDOCK221F00G	MicroDock II Docking Module for CO		1,495.00
BWTDOCK221G00G	MicroDock II Docking Module for SO ₂		1,495.00
BWTDOCK221H00G	MicroDock II Docking Module for O ₂		1,495.00



ToxiRAE Pro Series Monitors



Available in 3 different versions: for toxic gas sensors, with an LEL sensor or a PID. All instruments in the series share the same overall design, user interface, software, spare parts and accessories, providing significant savings on the cost of maintenance and training for plant wide deployments. Rechargeable internal lithium ion batteries provide up to 20 hours of runtime for toxic versions, and 12 hours for the LEL and PID models. The datalogging option can store 3 months of continuous data at an interval of 1 reading per minute. A best in class large, graphical, LCD display has a white backlight and provides gas levels and system status. A rugged and resilient enclosure protects from dust and water, and every monitor is rated intrinsically safe. Toxic and LEL versions are IP65 rated, and the PID version is IP54 rated.

Alarm notifications are provided by a 5 way system. Monitors feature built-in bright red LEDs, a 95 dB audible alarm, on screen notification with backlight activation, and a vibration alarm.

Each detector features a replaceable (rechargeable) battery, gas filter and fan (on PID models), easily replaceable sensors, and an alligator clip for easy attachment to clothing or belts. A rubberized boot adds protection from falls and wear and tear. A charging cradle and PC communications link are also included so data can easily be retrieved. ProRAE Studio software (included) helps to organize and store data for OSHA compliance.

ToxiRAE Pro: Can be equipped with 1 of over 20 different electrochemical gas sensors. Smart sensors store their own calibration data so sensors can be swapped out without having to recalibrate.

ToxiRAE Pro LEL: Measures the lower explosive limit of combustible gases with a range of 0 to 100%. Features 34 built in correction factors for specific gases.

ToxiRAE Pro PID: Provides a choice of 2 different PID sensors, a standard sensor with a 1 to 1,000 ppm (1 ppm resolution) range, or a high end PID sensor with a 0.1 to 2,000 ppm (0.1 ppm resolution) range. The high end PID is also available with a 9.8 eV lamp instead of the standard 10.6 eV. All ToxiRAE Pro PID monitors feature a 190 gas correction factor library.

Automatic bump testing and sensor calibration are available with the AutoRAE 2 unit. Contact Total Safety for more information.

Wireless capable monitors available. Contact Total Safety for more information.

TOXIC GASES

Cat. No.	Description	Purchase Price
RAEG02BE10100	Monitor ¹ , O ₂	\$399.00
RAEG02B210100	Monitor ¹ , CO ⁵	399.00
RAEG02BD10100	Monitor ¹ , CO High Range ⁷	499.00
RAEG02BF10100	Monitor ¹ , CO ⁸	599.00
RAEG02B110100	Monitor ¹ , H ₂ S ⁴	399.00
RAEG02BN10100	Monitor ¹ , H ₂ S High Range ⁷	399.00
RAEG02B310100	Monitor ¹ , SO ₂	499.00
RAEG02B410100	Monitor ¹ , NO	499.00
RAEG02B510100	Monitor ¹ , NO ₂	499.00
RAEG02B610100	Monitor ¹ , Cl ₂	499.00
RAEG02B710100	Monitor ¹ , HCN	499.00
RAEG02B810100	Monitor ¹ , NH ₃	499.00
RAEG02B910100	Monitor ¹ , PH ₃ ²³	499.00
RAEG02BP10100	Monitor ¹ , PH ₃ High Range ⁷	599.00
RAEG02BA10100	Monitor ¹ , ClO ₂	599.00
RAEG02BH10100	Monitor ¹ , CH ₂ -SH	599.00
RAEG02BJ10100	Monitor ¹ , EtO ⁴	499.00
RAEG02BK10100	Monitor ¹ , EtO ²	499.00
RAEG02BL10100	Monitor ¹ , EtO ³	599.00
RAEG02BM10100	Monitor ¹ , H ₂ S	499.00

¹ Includes ToxiRAE Pro gas monitor, specified sensor (installed), rechargeable internal lithium ion battery (installed), alligator clip (installed), protective rubber boot (installed), charging/PC connection dock, PC connection cable, AC adapter, calibration adapter, quick start guide, user manual on CD-ROM, ProRAE Studio II software, calibration/test certificate, and a warranty/registration card

² 0-10 ppm, ³ 0-10 ppm, ⁴ 0-100 ppm, ⁵ 0-500 ppm, ⁶ 0-1,000 ppm, ⁷ 0-2,000 ppm, ⁸ H₂ compensated

WITH PID

Cat. No.	Description	Purchase Price
RAEG02B010000	PID ¹ , Datalogging, 10.6 eV Standard PID ²	\$999.00
RAEG02B000000	PID ¹ , Datalogging, 10.6 eV High End PID ³	1,699.00

¹ Includes ToxiRAE Pro PID monitor, specified PID sensor (installed), internal rechargeable lithium ion battery (installed), alligator clip (installed), protective rubber boot (installed), charging/PC connection dock, PC connection cable, AC adapter, calibration adapter, sensor cap removal tool, quick start guide, user manual on CD-ROM, ProRAE Studio II software, calibration/test certificate, and warranty/registration card

² Standard PID, 1 to 1,000 ppm, 1 ppm resolution

³ High end PID, 0.1 to 2,000 ppm, 0.1 ppm resolution, includes documentation with correction factors and energy levels for over 300 VOCs

WITH LEL

Cat. No.	Description	Purchase Price
RAEG02B030000	LEL Monitor ¹ , Datalogging	\$499.00

¹ Includes: ToxiRAE Pro LEL monitor, LEL sensor (installed), internal rechargeable lithium ion battery (installed), alligator clip (installed), protective rubber boot (installed), charging/PC connection dock, PC connection cable, AC adapter, calibration adapter, quick start guide, user manual on CD-ROM, ProRAE Studio II software, calibration/test certificate, and a warranty/registration card



ToxiRAE 3 Monitor

Full-featured single gas monitor for H₂S and CO. Available features include 0.1 ppm H₂S resolution and high-range 1999 ppm CO, event logging of up to 10 events and 2-point calibration for 0 and span. Comes standard with 2/3 AA high-capacity lithium batteries, audible, visual and vibrating alarms, calibration adapter and stainless-steel IP67-rated housing for its water- and dust-resistance. Also highly resistant to RFI and EMI interference. AutoRAE Lite compatible (see page 116).

2-year warranty including sensor and battery if used within specifications.

UL approved and CSA certified for Class I, Division 1, Groups A, B, C and D; T4.



INDUSTRIAL SCIENTIFIC

GasBadge® Pro Dockable Single Gas Monitor

Interchangeable "smart" sensors monitor unsafe levels of oxygen or toxic gases.

Communicates via an infrared interface to optional accessories like the DS2 Docking Station™ (see page 114) or Datalink and infrared printer to further simplify and automate calibration, function (bump) testing and data downloading. Standard STEL and TWA readings, and datalogging up to 1 year of survey data are featured along with an event logger that records the last 15 alarm events.

Monitor's housing is immune to RF water-resistant and extremely durable. Concussion-proof overmold protects the unit from extreme abuse. Simple and intuitive 4-button navigation allows easy access to setup, operation and calibration functions via the backlit LCD display. Comes standard with Low and High visual, audible and vibrating alarms. Parts-per-million or percent-by-volume readout. Replaceable CR2 lithium battery. Lifetime warranty.

UL approved and CSA certified for Class I, Division 1, Groups A, B, C and D.



Cat. No.	Description	Rental	Purchase Price
RAEG010102000	H ₂ S	YES	\$185.00
RAEG010101000	CO (0 to 500 ppm)	YES	185.00
RAEG010103000	CO (0 to 1999 ppm)		185.00

ACCESSORIES

Cat. No.	Description	Rental	Purchase Price
RAET010100000	AutoRAE Lite Bump Test and Calibration Station	YES	\$1,231.00

Cat. No.	Description	Rental	Purchase Price
ISC181000601	CO	YES	\$375.00
ISC181000602	H ₂ S	YES	375.00
ISC181000603	O ₂		375.00
ISC181000604	NO ₂		525.00
ISC181000605	SO ₂	YES	525.00
ISC181000606	NH ₃	YES	595.00
ISC181000607	Cl ₂	YES	525.00
ISC181000608	ClO ₂	YES	595.00
ISC181000609	PH ₃		645.00
ISC18100060B	HCN	YES	595.00
ISC18100060C	H ₂		525.00
ISC18100060G	CO/Low H ₂ Interference		525.00

ACCESSORIES

Cat. No.	Description	Purchase Price
ISC18106484	Nylon Carrying Case	\$11.00
ISC18106492	2-Unit Nylon Carrying Case	16.50
ISC18106302ABC*	DS2 Docking Station for GasBadge Pro	Call for Quote
ISC18106500	Constant-Flow Hand Aspirated Pump	195.00
ISC18106260	Datalink	165.00

*All ABC extensions require:
A = Wireless Option
B = Number of iGas Readers
C = Power Cord Option



Trust Total Safety to deliver the industry's best gas detection equipment where you can expect to:

- Rent premium equipment at reasonable prices
- Optimize your investment in safety equipment
- Get first class quality of service and repair
- Utilize testing and calibration support
- Minimize bottlenecks during turnarounds



INDUSTRIAL SCIENTIFIC

GasBadge® Plus Personal Single Gas Monitor



1-year continuous monitoring, maintenance-free, dockable single gas monitor ideal for personal protection from unsafe levels of O₂, CO, H₂S, SO₂ or NO₂. Compact and lightweight (2½ oz – 72 grams), monitor can be worn comfortably to provide continuous and unobstructed protection. Third-party certified IP66/67 to be extremely durable and resistant to water and radio frequency interference. Concussion-proof overmold protects from extreme abuse in a variety of harsh industrial environments.

Standard features include a powerful audible Low and High alarms complemented by vibrating and visual alarms and a non-replaceable lithium-ion battery. Backlit LCD provides readout in PPM or percent-by-volume. DS2 Docking Station™ (see page 114) compatible.

1-year warranty.

Cat. No.	Description	Rental	Purchase Price
ISC181000501	CO	YES	\$195.00
ISC181000502	H ₂ S	YES	195.00
ISC181000503	O ₂		195.00
ISC181000504	NO ₂		325.00
ISC181000505	SO ₂	YES	325.00

ACCESSORIES

Cat. No.	Description	Purchase Price
ISC18106401	Nylon Carrying Case	\$11.00
ISC18106419	2-Unit Nylon Carrying Case	16.50
ISC17121963	Neck Lanyard w/Safety Release	5.50
ISC18106500	Constant-Flow Hand Pump	195.00
ISC17124504	Replacement Water/Dust Sensor Barriers, 5 Count	11.00
ISC18106260	Datalink	165.00
ISC1810634400	Cal Plus Calibration Station	980.00
ISC1810634410	Cal Plus Calibration Station w/On-Board Printer	1,350.00
ISC17117722	Serial Data Dot Matrix Printer, 120V AC Powered	527.90
ISC18107698ABC*	DS2 Docking Station for GasBadge Plus	Call for Quote

*All ABC extensions require:

A = Wireless Option

B = Number of iGas Readers

C = Power Cord Option

Honeywell ToxiPro® Monitors



Economical, full-featured, single gas monitors. The compact and rugged monitors are rated to IP65/IP67 against water ingress and dust. Each easy-to-use monitor features an overmolded boot, one-button operation, automatic calibration and a large display. Equipped with IrDA® PC communication, the monitors feature an automatic event logger up to 20 hours. Standard accessories include a heavy-duty alligator clip, calibration adapter and reference manual.

Reusable monitor with a replaceable lithium battery (9000 hours of operation) comes standard with a black-box recorder and optional datalogging capabilities. Includes backlit display and features

Current, Max, STEL and TWA readings and Low, High, STEL and TWA audible and visual alarms. Optional vibrating alarm.

Cat. No.	Description	Rental	Purchase Price
SPR544590	O ₂		\$234.00
SPR544501	CO	YES	244.00
SPR544519	CO – H (Reduced Sensitivity to H ₂)		304.00
SPR544502	H ₂ S		244.00
SPR544503	SO ₂		421.00
SPR544521	NH ₃		404.00
SPR544518	Cl ₂		380.00
SPR544509	NO ₂		380.00
SPR544510	HCN		380.00
SPR544513	PH ₃		380.00

IQ Express™ Docking Station

Drop the ToxiPro detector into this fully automated bump test and calibration station to perform up to 8 critical tests. Features automatic full span calibration in as little as 20 seconds. See page 117 for more information.





LEL/UEL for Gases Commonly Used in Industry

A hazardous vapor concentration is present when a gas reaches a level known as the Lower Explosive Limit (LEL). Gases below the LEL are considered too lean to burn. If the concentration exceeds the Upper Explosive Limit (UEL), the gas is considered too rich to burn. A gas concentration between these two limits is considered to be in its flammable range and will ignite and burn if exposed to an ignition source. Avoid unwanted fires and explosions by keeping gas concentration below the LEL, thus preventing it from reaching its flammable range.

Substance	LEL (%Vol of Air)	UEL (%Vol of Air)
Acetic Acid	4.0%	19.9% (200°F [93°C])
Acrylic Acid	2.4%	8.0%
Ammonia	15.0%	28.0%
Benzene	1.2%	7.8%
1,3-Butadiene	2.0%	12.0%
n-Butane	1.6%	8.4%
2-Butanone	1.4% (200°F [93°C])	11.4% (200°F [93°C])
Carbon Monoxide	12.5%	74.0%
Ethylene Oxide	3.0%	100.0%
Ethylene Dichloride (EDC)	6.2%	16.0%
Formaldehyde	7.0%	73.0%
Gasoline	1.4%	7.6%
n-Hexane	1.1%	7.5%
Hydrogen Cyanide	5.6%	40.0%
Hydrogen Sulfide	4.0%	44.0%
Methyl Alcohol	6.0%	36.0%
Methyl Chloride	8.1%	17.4%
Styrene	0.9%	6.8%
Toluene	1.1%	7.1%
Turpentine	0.8%	?
Vinyl Chloride	3.6%	33.0%
Xylene, m, p	1.1%	7.0%
Xylene, o	0.9%	6.7%

Source: www.cdc.gov/niosh/npg
NIOSH Pocket Guide to Chemical Hazards

3M Personal Air Monitoring System

Designed to monitor personnel's exposure to possible contaminants. The exposure is determined by means of diffusion sampling and is then compared to the industry standard to determine whether the exposure presents health hazards.

Can be purchased as a monitor only or with the 3M™ analysis service. To use, simply attach the badge near the worker's breathing zone for 8 hours, and return it to 3M for confidential written analysis to verify exposure to a TWA for OSHA requirements.

Monitor has an 18-month shelf life and meets accuracy of $\pm 25\%$ for many workplace contaminants.



Formaldehyde Monitors

Monitors personnel or work areas in health care, laboratories and in the chemical, pulp/paper, foundry and textile industries. Can be used for 8-hour TWA samplings to meet the OSHA requirement of $\pm 25\%$ accuracy at 0.5 to 5.0 ppm formaldehyde. However, these monitors do not have the sensitivity for STEL monitoring.

Cat. No.	Description	Purchase Price
MMM3721	Monitor, Case of 5	\$165.78
MMM3720	Monitor with Prepaid Analysis, Case of 5	413.59
MMM3721B	Monitor, Case of 25	768.00

Ethylene Oxide Monitors

Designed for sampling personnel or work areas in the pharmaceutical, health care and chemical industries. Monitors utilize the proven scientific principle of diffusion, which is the gradual mixing of molecules of two or more substances. Meet OSHA's requirement for accuracy within $\pm 25\%$ for 8 hour samplings at 1 ppm and the requirement for accuracy within $\pm 35\%$ for 8 hour samplings at the 0.5 ppm action limit.

Cat. No.	Description	Purchase Price
MMM3550	Monitor with Prepaid Analysis, Case of 5	\$371.67
MMM3551	Monitor, Case of 5	132.62

Organic Vapor Monitors

Designed for sampling personnel exposure to organic vapors in industrial work areas. The MMM3500 and MMM3510 utilize a single sorbent pad to collect organic vapors, while the MMM3520 and MMM3530 include a back-up pad for sampling higher concentrations or vapors that don't adsorb as well onto carbon. Each monitor can be analyzed for up to 3 vapors from a list of more than 125 organic compounds. Call Total Safety for a complete listing.

Cat. No.	Description	Purchase Price
MMM3510	Monitor with Prepaid Analysis, Case of 5	\$618.76
MMM3530	Monitor with Backup Pad and Prepaid Analysis, Case of 5	835.08



Rig Rat III Portable Area Monitoring System



This self-powered and solar-capable system provides multi-point monitoring for toxic gases, combustibles and oxygen hazards with a variety of field configurations. Trenching and cabling costs are eliminated through wireless, two-way digital RF data transmission.

License-free, 2.4 GHz band RF transceiver provides 2-way, real-time communication with the controller located up to 1.8 miles (2.9 km) away. Encoded digital signal is RFI-protected. Standard package uses a 200 mW radio. 10 mW radio is available.

Monitor contains an intrinsically safe, weatherproof, stainless-steel enclosure. User can add, change or expand the system to include up to several hundred individual monitoring points, which may require more than one controller. Features a LCD display, a simple push-button operation for automatic calibration and setup choices, visual and audible alarms, and battery-powered, solar and line-capable power options.

Detector can be equipped with one or 2 remote gas sensor heads in any gas combination. A simple rotary switch is used to choose the sensor type and the measuring range. Alarm points can be field-assigned.

Monitors come standard with a battery back-up, 5 DBI antenna, 2 remote sensor ports, 2 option ports and a power supply port. Contact Total Safety for distances over 0.6 mile (.9 km).

2-year warranty including sensors. 1-year warranty for ammonia.

Cat. No.	Description	Rental	Purchase Price
BWTRR3000	Wireless Multi-Point Systems	YES	\$2,680.00
BWTRR3000S	Non-Wireless Version		2,680.00

REMOTE SENSOR UNITS

Cat. No.	Description	Rental	Purchase Price
BWTSAIL3H2	H ₂ S (0-50 ppm)		\$425.00
BWTSAIL3H1	H ₂ S (0-100 ppm)	YES	425.00
BWTSAIL3H3	H ₂ S (0-500 ppm)	YES	425.00
BWTSAIL3W1	LEL	YES	475.00
BWTSAIL3S3	SO ₂ (0-20 ppm)		550.00
BWTSAIL3S1	SO ₂ (0-100 ppm)		550.00
BWTSAIL3A2	NH ₃ (0-100 ppm)		795.00
BWTSAIL3D1	NO ₂ (0-10.0 ppm)		895.00
BWTSAIL3X1	O ₂ (0-25% Vol)		425.00
BWTSAIL3X2	O ₂ (0-30% Vol)		425.00
BWTSAIL3M1	CO (0-500 ppm)	YES	450.00

INTRINSICALLY SAFE BARRIERS

Cat. No.	Description	Purchase Price
BWTRRSBAR1	Sensor Barrier (Required for Group B Applications)	\$485.00

Contact Total Safety for additional accessories and options.



Total Safety's exclusive In-Plant Service Centers (IPSCs) allows us to be onsite, day and night, with full-time, trained technicians providing integrated safety services such as:

- Planning of safety practices
- Supply of rental equipment
- Inspection and maintenance of equipment
- Distribution of safety instruments
- Training
- Gas, fire and respiratory inspection and compliance



AreaRAE Steel Portable Area Monitoring Detector

• Stainless-Steel Housing

A 1- to 5-sensor gas detector equipped with a wireless RF modem for communication and transmission of readings and other information on a real-time basis with a remote base controller located up to 2 miles (3.2 km) away. A PC can be used as the controller and standard ProRAE Remote software can monitor input from up to 32 locations.

First "lunchbox"- type multi-sensor instrument able to include a PID for parts-per-million measurement of VOCs, as well as LEL, O₂ and up to 2 electrochemical sensors to measure specific toxic substances such as CO and H₂S.

In stand-alone operation, the weather-resistant, portable stainless-steel monitor can run over 24 hours on either rechargeable lithium-ion or alkaline batteries. Includes carbon filters for units with CO sensor, calibration adapter, rechargeable lithium-ion battery, spare alkaline battery, 120/240V AC, AC/DC adapter, 10 spare external filters, hard transport case and 15-ft (4.6 m) Teflon® tubing. Optional GPS provides the ability to track and display readings from remote detectors.

Datalogging Monitors include ProRAE Suite Package for Windows® 98, Windows NT, Windows 2000 and Windows XP and a computer interface cable. ProRAE Remote Basic Host software package includes the ProRAE Remote Software and a RAELink modem.

Calibration Kits include detector, calibration gas and calibration regulator and tubing.

Standard Rapid Deployment Kit Packages include detector, ProRAE Remote package in military-grade case including laptop and RAELink2 modem (see page 104), 4 Monitors, RAELink2 Repeater, external 4-bank battery charger, 4 spare lithium-ion battery packs, 4 alkaline adapters and a SensorRAE.



GPS OPTION AND CALIBRATION KITS

Cat. No.	Description	Purchase Price
RAE039W110027	LEL, O ₂ , H ₂ S	Call for quote
RAE039W120026	LEL, O ₂ , CO	Call for quote
RAE039W100124	LEL, O ₂ , PID	Call for quote
RAE039W112121	LEL, O ₂ , H ₂ S, CO, PID	Call for quote

STANDARD RAPID DEPLOYMENT KIT PACKAGES

Cat. No.	Description	Purchase Price
RAE039R121111	LEL, O ₂ , H ₂ S, CO, PID	Call for quote
RAE039R121121	LEL, O ₂ , H ₂ S, CO, PID with GPS Option	Call for quote

AreaRAE Host Controller

Base station for networks based on ProRAE Remote base-station software. Integrates the output of up to 32 remote sensors located up to 2 miles (3.2 km) away on the user display.

Rapid Deployment Kit Host Kits include ProRAE Remote Package in military-grade case including laptop, RAELink2 Modem (see page 104), 12-in antenna cable and charging to control up to 32 wireless detectors.

Cat. No.	Description	Rental	Purchase Price
RAE039S100000	RDKit Host Kit	YES	Call for quote
RAE039S200000	RDKit Host Kit with GPS Option		Call for quote
RAE0390505000	RAELink3		Call for quote

AreaRAE Multigas Monitor

• Polycarbonate Housing

Cat. No.	Description	Rental
RAE039S100000	LEL, O ₂ , H ₂ S	YES
RAE039S200000	LEL, O ₂ , CO	YES
RAE0390502000	LEL, O ₂ , PID	YES
RAE039W112010	LEL, O ₂ , H ₂ S, CO	YES
RAE039W112110	LEL, O ₂ , H ₂ S, CO, PID	YES

For a complete listing of available configurations, please contact Total Safety.

CALIBRATION KITS

Cat. No.	Description	Rental	Purchase Price
RAE039W110017	LEL, O ₂ , H ₂ S	YES	Call for quote
RAE039W120016	LEL, O ₂ , CO	YES	Call for quote
RAE039W100114	LEL, O ₂ , PID	YES	Call for quote
RAE039W112015	LEL, O ₂ , H ₂ S, CO	YES	Call for quote
RAE039W112111	LEL, O ₂ , H ₂ S, CO, PID	YES	Call for quote



MeshGuard Single Gas Detector

• For Semi-fixed Area Monitoring

A wireless, single-toxic-gas detector with an integrated wireless mesh radio modem ideal for semi-fixed gas detection on hazardous safety projects lasting from a week to a year. Creates a network by routing data around obstacles and providing redundant routing with an operating distance up to 1,000 ft (300 m) between nodes. Housing is rated IP65 for water resistance and is RFI and EMI protected. Monitor features a backlit LCD display for ease of use and High, Low, STEL and TWA visual and audible alarms. Vibrating alarm optional.

Remote readings are displayed via the FMC-2000 wireless controller, which can support up to 24 detectors or via

a PC controller, which can support up to 128 detectors. Reports real-time information to portable controller or PC-based software and can be deployed in under 10. Standard 2-point field calibration.

Monitor comes standard with calibration gas test adaptor, replaceable lithium battery and steel bracket for mounting. Magnet mount optional.

Extension Kits come standard with choice of 6 monitors (choice of H₂S, CO, LEL or any combination), transport case, mounting bracket with pole mount, magnet mount, battery, 10 sensor filters, calibration cap, O-rings and an antenna.

Compliance Kits come with similar products as the Extension Kits but also include a FMC-2000 controller.

UL approved and CSA certified as intrinsically safe for use Class I, Division 1, Groups A, B, C and D; T4. Contact Total Safety for country-specific certifications.

Cat. No.	Description	Rental	Purchase Price
RAED01B001011	H ₂ S (0.1 to 100 ppm)	YES	Call for Quote
RAED01B00G011	LEL (1 to 100% LEL)	YES	Call for Quote
RAED01B003011	CO (1 to 500 ppm)	YES	Call for Quote
RAED01B00D011	O ₂ (0 to 25% Vol.)	YES	Call for Quote
RAED01B00A011	NH ₃ (0 to 100 ppm)	YES	Call for Quote
RAED01B008011	Cl ₂ (0 to 10 ppm)	YES	Call for Quote
RAED01B005011	SO ₂ (0.1 to 20 ppm)	YES	Call for Quote

KITS

Cat. No.	Description	Rental	Purchase Price
RAED01R044203	H ₂ S Extension Kit (without Controller)	YES	Call for Quote
RAED01R144203	H ₂ S Compliance Kit (with Controller)	YES	Call for Quote

ACCESSORIES

Cat. No.	Description	Purchase Price
RAEF010901000	FMC-2000 Controller Kit	Call for Quote
RAED01A000010	Router	Call for Quote

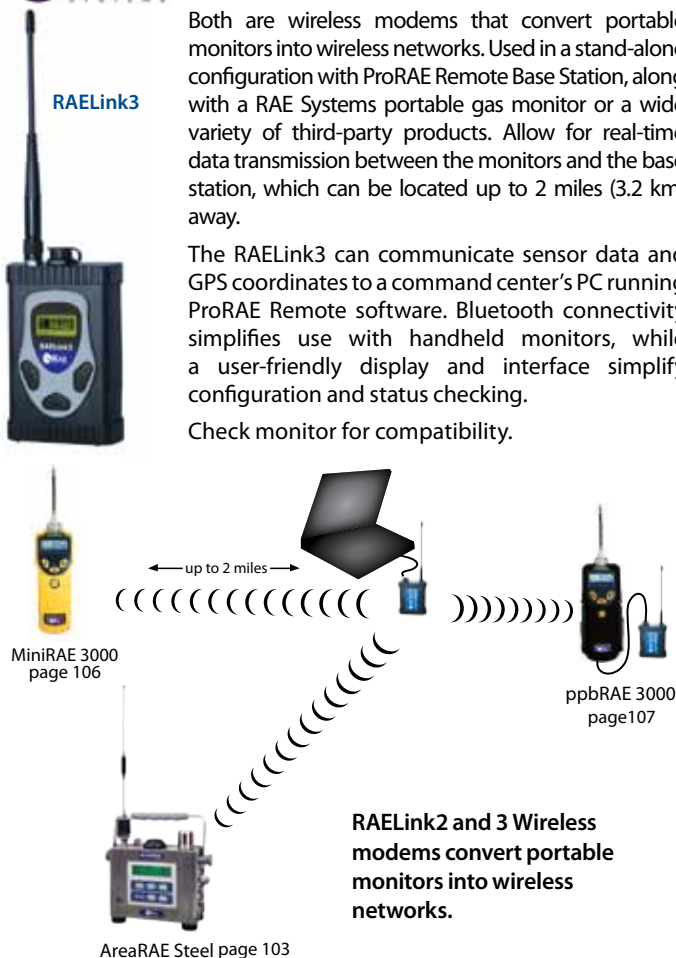


RAELink2 and 3 Wireless Modems

Both are wireless modems that convert portable monitors into wireless networks. Used in a stand-alone configuration with ProRAE Remote Base Station, along with a RAE Systems portable gas monitor or a wide variety of third-party products. Allow for real-time data transmission between the monitors and the base station, which can be located up to 2 miles (3.2 km) away.

The RAELink3 can communicate sensor data and GPS coordinates to a command center's PC running ProRAE Remote software. Bluetooth connectivity simplifies use with handheld monitors, while a user-friendly display and interface simplifies configuration and status checking.

Check monitor for compatibility.



Cat. No.	Description	Rental	Purchase Price
RAE0295001200	RAELink2 for Use with Base Station (Includes ProRAE Remote Software)	YES	Call for Quote
RAE0290505000	RAELink3 for Use with Portable Monitors	YES	Call for Quote



EchoView Host Wireless Mini-Controller for Wireless Portable Monitors

Rugged handheld wireless device can establish a self-contained, self-sufficient network with up to 8 supported RAE systems portable wireless monitors and display their readings and alarm status in real time on an easy-to-read screen. Mesh routers extend the communications range between the EchoView Host and supported portable gas monitors.

Cat. No.	Description	Rental	Purchase Price
RAEF040005000	EchoView Host Mini-Controller, 915 MHz	YES	Call for Quote
RAEF040008000	Mesh Router, 915MHz	YES	Call for Quote
RAE500011100001	Replacement Battery for EchoView or Mesh Router, 3.6V (each)		Call for Quote
RAE500011100005	Replacement Battery for EchoView or Mesh Router, 3.6V (pack of 5)		Call for Quote



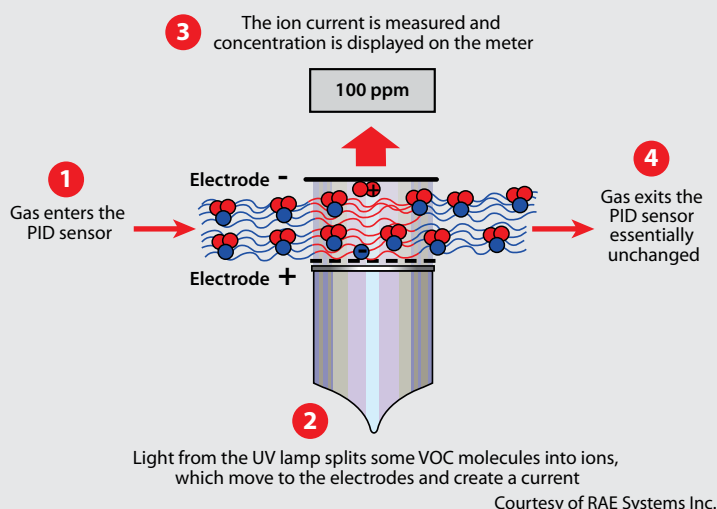
PID Technology Brings Benzene (and other VOC) Detection to a New Level

Benzene is a natural constituent of crude oil, an important industrial solvent, and a precursor commonly used in the production of drugs, plastics, synthetic rubber, and dyes.

Benzene is also a known carcinogen and can also cause other long term health effects. For this reason, it is important for workers to monitor benzene levels for entry pre-screening during refinery and plant maintenance, hazardous material response, marine spill response and refinery downstream monitoring.

A photoionization detector (PID) offers a low-cost technology to protect workers not only from benzene levels, but other volatile organic compounds (VOCs) and hydrocarbons (HCs). PIDs provide great sensitivity and fast response at extremely low concentration of compounds (parts per billion).

How a PID Works



Benzene Specifications

Formula: C_6H_6

CAS: 71-43-2

Synonyms & Trade Names: Benzol, Phenyl Hydride

Exposure Limits:

NIOSH REL: Ca, TWA 0.1 ppm, ST 1 ppm

OSHA PEL: [1910.1028] TWA 1 ppm, ST 5 ppm

IDLH: Ca [500 ppm]

Conversion: 1 ppm = 3.19 mg/m³

Physical Description:

MW: 78.1 BP: 176°F (80°C) FRZ: 42°F (5.5°C) Sp.Gr: 0.88

VP: 75 mmHg IE (IP): 9.24 eV FL.P: 12°F (11°C) Sol: 0.07%

LEL: 1.2% UEL: 7.8%

Incompatibilities & Reactivities: Strong oxidizers, many fluorides & perchlorates, nitric acid

Measurement Methods: NIOSH 1500, 1501, 3700, 3800; OSHA 12, 1005

Correction Factors & Ionization Energies for PIDs

In general, any compound with ionization energy (IE) lower than that of the lamp photons can be measured. The best way to calibrate a PID to different compounds is to use a standard of the gas of interest. However, correction factors have been determined that enable the user to quantify a large number of chemicals using only a single calibration gas, typically isobutylene.

Compound Name	CF for Lamp			IE eV	TWA
	9.8 eV	10.6 eV	11.7 eV		
Acetic Acid	NR	22	2.6	10.66	10
Acrylic Acid		12	2.0	10.60	2
Ammonia	NR	9.7	5.7	10.16	25
Benzene	0.55	0.53	0.6	9.25	0.5
Bromine	NR	1.30	0.74	10.51	0.1
Butadiene	0.8	0.85	1.1	9.07	2
Butane		67	1.2	10.53	800
Ethane		NR	15	11.52	ne
Ethylene Oxide		13	3.5	10.57	1
Formaldehyde	NR	NR	1.6	10.87	C0.3
Gasoline #1		0.9			300
Gasoline #2, 92 Octane	1.3	1.0	0.5		300
Hexane, n-	350	4.3	0.54	10.13	50
Hydrogen Cyanide	NR	NR	NR	13.6	C4.7
Hydrogen Sulfide	NR	3.3	1.5	10.45	10
Jet Fuel, JP-4		1.0	0.4		ne
Jet Fuel, JP-5		0.6	0.5		29
Jet Fuel, JP-8		0.6	0.3		30
Methane	NR	NR	NR	12.61	ne
Methanol	NR	NR	2.5	10.85	200
Methyl Chloride	NR	NR	0.74	11.22	50
Methyl Ethyl Ketone	0.86	0.9	1.1	9.51	200
Nitric Oxide	~6	5.2	2.8	9.26	25
Nitrogen Dioxide	23	16	6	9.75	3
Styrene	0.45	0.40	0.4	8.43	20
Tetrafluoroethliene		~15		10.12	ne
Toluene	0.54	0.5	0.51	8.82	50
Turpentine	0.37	0.3	0.29	~8	20
Vinyl Chloride		2.0	0.6	9.99	5
Xylene, m-	0.50	0.44	0.40	8.56	100
Xylene, o-	0.56	0.46	0.43	8.56	100
Xylene, p-	0.48	0.39	0.38	8.44	100

NR= No Response

ne= Not Established ACGIH 8-hr TWA

IE= Ionization Energy

C##= Ceiling value, given where 8-hr TWA is not available

Courtesy of RAE Systems, Inc.

RAE Systems Benzene-Specific PIDs

UltraRAE 3000

See page 106 for details





UltraRAE 3000 Benzene Monitor



Compound-specific monitor has a range of 0.05 to 10,000 ppm in VOC mode and 0.05 ppm to 200 ppm in benzene-specific mode. Features a 3-sec response time in VOC mode and a 60-sec response time in compound-specific mode, automatic temperature-controlled sampling time calculation and wireless data transmission with built-in Bluetooth. RAELink3 (see page 104) portable modem optional.

Monitor comes standard with an integrated RAE Systems Correction Factors list for more than 200 compounds, a rechargeable lithium-ion battery and interchangeable alkaline battery adapter and a large graphic display. Features a 15-min benzene STEL calculation with 15-min real-time benzene-specific measurement and recording. The highly specific readings are due to the combined 9.8 eV UV lamp and RAE-Sep™ benzene tube, which provide direct readings of benzene and butadiene. Sampling probe design provides instant tube-breakthrough visibility.

The housing of the monitor is IP65-rated with the unit running and is highly resistant to EMI and RFI. Audible and visual alarms detect High, Low, STEL and TWA readings. Datalogging capabilities are standard 6 months at 1-min intervals. Features 2-point or 3-point calibration for zero and span downloadable to a PC through charging cradle.

Standard accessories include datalogging with ProRAE Studio Package for Windows® NT, Windows 2000 and Windows XP, travel charger, Flex-I-Probe™ and short probe, external filter, red rubber boot, universal AC/DC charger and a soft leather carrying case.

Accessory Kits include standard accessories with a hard transport case with pre-cut foam padding, charging/download cradle (instead of charging/download travel adapter), 5 porous metal filters and O-rings, organic vapor zeroing kit and gas outlet port adapter and tubing.

Calibration Kits include standard accessories and accessory kit, as well as calibration gas (34 L) and a calibration regulator and flow controller.

3-year warranty for 10.6 eV lamp and 1-year warranty for 9.8 eV lamp, pump, battery, sensor and instrument.

UL approved and CSA certified as intrinsically safe for use in Class I, Division 1, Groups A, B, C and D.

Cat. No.	Description	Rental	Purchase Price
RAE059D310000	Unit with Universal Charger	YES	\$5,022.00
RAE059D311200	Unit with Universal Charger, Accessories Kit, Calibration Kit and Wireless Bluetooth	YES	5,846.00
RAE0123022010	RAE-Sep Benzene Tubes, Box of 10		72.00



MiniRAE 3000 VOC PID

- 0 to 15,000 ppm Range



Handheld monitor detects volatile organic compounds with an extended range of 0 to 15,000 ppm. The built-in RF modem allows for real-time data transmission with a base controller located up to 500 ft (152.4 m) away or up to 2 miles (3.2 km) away with optional RAELink3 portable modem (see page 104). A PC can be used as a base station. Standard ProRAE Remote software is capable of monitoring the input of up to 64 remotely located monitors. Wireless Bluetooth optional.

Monitor features a 3-sec response time, an LCD backlight, 12 encoded languages and an integrated RAE Systems Correction Factors list for more than 200 compounds for measurement. RFI- and EMI-resistant, housing is IP67-rated for its waterproof design and comes with a removable rubber boot. Features both visual and audible alarms for Low, High, STEL and TWA readings. Standard datalogging capabilities up to 6 months at 1-min intervals. Rechargeable lithium-ion battery pack is interchangeable with an alkaline battery pack. Standard charging adapter also doubles as a download adapter.

Standard accessories include datalogging with ProRAE Studio Package for Windows® 98, Windows NT, Windows ME, Windows 2000 and Windows XP, Flexi-Probe™ sample probe, built-in sample pump, external filter, lamp-cleaning kit, a tool kit and a soft leather case.

Accessory Kits include standard accessories as well as a hard transport case with pre-cut foam padding, 5 porous metal filters with O-rings, organic vapors zeroing kit and a gas outlet port adapter and tubing.

Calibration Kits include all standard accessories and accessories from Accessory Kits, as well as 100 ppm isobutylene calibration gas (34 L) and calibration regulator and flow controller.

3-year warranty for 10.6 eV lamp and 1-year warranty for pump, battery, sensor and instrument.

UL approved and CSA certified for Class I, Division 1, Groups A, B, C and D. Contact Total Safety for additional compliances.

Cat. No.	Description	Rental	Purchase Price
RAE059B110000	Unit with Standard Accessories	YES	\$3,940.00
RAE059B111100	Unit with Accessory Kit, Calibration Kit and Wireless Bluetooth		4,764.00



MiniRAE Lite VOC PID

• 0 to 5,000 ppm Range

Handheld monitor detects volatile organic compounds with an extended range of 0 to 5,000 ppm. Monitor features a 3-sec response time, an LCD backlight, 12 encoded languages and an integrated flashlight. Comes standard with Duty-cycling™ lamp and sensor auto-cleaning technology, a tough, flexible inlet Flexi-Probe™ and a built-in sample pump.

Housing is IP65-rated for its waterproof design and comes with a removable rubber boot. Features both visual and audible alarms for Low, High, STEL and TWA readings. Rechargeable lithium-ion battery pack is interchangeable with an alkaline battery pack.

Standard accessories include Flexi-Probe™ sample probe, external filter, green rubber boot, alkaline battery adapter, lamp cleaning kit, tool kit and a soft leather carrying case.

Accessory Kits include standard accessories, as well as a hard transport case with pre-cut foam padding, 5 porous metal filters with O-rings, organic vapors zeroing kit and a gas outlet port adapter and tubing.

Calibration Kits include all standard accessories and accessories from Accessory Kits, as well as 100 ppm isobutylene calibration gas (34 L) and calibration regulator and flow controller.

3-year warranty for 10.6 eV lamp and 1-year warranty for pump, battery, sensor and instrument.



ppbRAE 3000 VOC PID

• 1 ppb to 10,000 ppm Range

Handheld monitor has an extended range of 1 ppb to 10,000 ppm. Built-in RF modem allows real-time data transmission with a base controller located up to 500 ft (152.4 m) away or 2 miles (3.2 km) away with optional RAElink3 portable modem (see page 104). A PC can also be used as the base station.

Monitor features a 3-sec response time, an integrated RAE Systems Correction Factors list with over 350 compounds and 10 encoded languages. Housing is IP65-rated for its waterproof design while running and IP67-rated while off and without inlet probe, as well as highly resistant to EMI and RFI. Monitor includes both audible and visual alarms for Low, High, STEL and TWA readings.

Comes standard with a large graphic display, interchangeable battery pack, integrated flash light, inlet Flex-I-Probe™, built-in sample pump, Duty-cycling™ lamp and sensor auto-cleaning technology, and datalogging capabilities up to 6 months at 1-min intervals. Features 2- or 3-point calibration for zero and span. AutoRAE calibration station compatible. Contact Total Safety for more information.

Standard accessories include datalogging with ProRAE Studio Package, black rubber boot with straps, external filter, rechargeable lithium-ion battery with universal AC/DC adapter, alkaline battery adapter, VOC zeroing tube (1 box), tube adapter, Tedlar® bag, travel download charger and soft leather carrying case.

Accessory Kits include standard accessories, as well as hard transport case with pre-cut foam padding, charging/download cradle, 5 porous metal filters and O-rings, organic vapor zeroing kit and gas outlet port adapter and tubing.

Calibration Kits include standard accessories and accessories from kits as well as 10 ppm isobutylene calibration gas (34 L) and calibration regulator and flow controller.

3-year warranty on 10.6 eV lamp and 1-year warranty for pump, battery, sensor and instrument.

UL approved and CSA certified as intrinsically safe for use in Class I, Division 1, Groups A, B, C and D.



Cat. No.	Description	Rental	Purchase Price
RAE059A110000	Unit Only	YES	\$2,570.00
RAE059A111100	Unit with Accessory Kit, Calibration Kit and Universal Charger		3,085.00



At Total Safety, we meticulously service all of the equipment we sell and have the industry expertise to facilitate the repair and maintenance for all popular gas detection products.

Cat. No.	Description	Purchase Price
RAE059C110000	Unit with Universal Charger	\$6,515.00
RAE059C111100	Unit with Universal Charger, Accessories Kit, Calibration Kit and Wireless Bluetooth	7,339.00



Kwik-Draw® Sampling Pump & Tubes



Can be used with an assortment of MSA's detector tubes to spot-test atmospheres for a wide variety of toxic substances. Designed for one-hand operation and consistent delivery of a sample draw volume of 100 mL. Constructed with a shaft-guided compression system for a more consistent and replicable flow rate and volume per stroke. Integral, easy-to-read stroke counter shows the exact number of strokes performed and provides a positive stop when the stroke has been completely compressed.

The Kwik-Draw® Deluxe Pump has a unique end-of-stroke indicator that "winks" after a precise amount of air is drawn confirming enough air has been sampled for a successful reading. Pump includes remote sampling adaptor and a hand-carrying pouch.

KWIK-DRAW® SAMPLING PUMPS

Cat. No.	Description	Rental	Purchase Price
MSA488543	Basic Pump	YES	\$530.00
MSA487500	Deluxe Pump		591.00

ACCESSORIES

Cat. No.	Description	Rental	Purchase Price
MSA73067	Sampling Line, 10 ft (3.1 m)	YES	\$55.00
MSA73069	Sampling Line, 50 ft (15.2 m)	YES	293.00
MSA87970	Remote Sampling Adaptor (Required for Sampling Line)	YES	11.00

DETECTOR TUBES

Cat. No.	Gas/Vapor	Measuring Range (ppm)	TLV (ppm)	Purchase Price
MSA804138	Acetic Acid	1-80	10	\$99.00
MSA804141	Acetone	100-10,000	500	99.00
MSA804405	Ammonia	2-500	25	102.00
MSA800300	Ammonia	20-1000	25	89.00
MSA804406	Ammonia	0.1 - 10 Vol %	25	89.00
MSA804132	Benzene	5-500	0.5	103.00
MSA804411	Benzene	5-100	0.5	89.00
MSA804136	n-Butyl Alcohol	100-3900	50 (ceiling)	89.00
MSA497606	Carbon Dioxide	100-3000	5000	99.00
MSA487333	Carbon Dioxide	0.1-7.0 Vol %	5000	89.00
MSA804419	Carbon Dioxide	1-20 Vol %	5000	89.00
MSA492514	Carbon Disulfide	2-300	10	89.00
MSA804421	Carbon Monoxide	0.001-0.3 Vol %	25	89.00

Cat. No.	Gas/Vapor	Measuring Range (ppm)	TLV (ppm)	Purchase Price
MSA803943	Carbon Monoxide	5-1000	25	\$89.00
MSA815507	Carbon Monoxide	3000-70,000	25	89.00
MSA804423	Carbon Monoxide	0.1-1.0 Vol %	25	71.00
MSA487335	Carbon Monoxide	0.3-7.0 Vol %	25	89.00
MSA803944	Chlorine	0.2-30	0.5	102.00
MSA804133	Chlorine Dioxide	0.05-15	0.1	102.00
MSA804132	Chlorobenzene	40-610	10	103.00
MSA804416	Dichloromethane	50-1000	50	99.00
MSA804136	Ethanol	100-6000	1000	89.00
MSA804428	Ethylene	50-5000	N/A	99.00
MSA497649	Formaldehyde	0.1-55	0.3 (ceiling)	99.00
MSA492870	Gasoline	30-6000	300	89.00
MSA803948	Hydrogen Chloride	1-30	5 (ceiling)	115.00
MSA803945	Hydrogen Cyanide	2-50	4.7 (ceiling)	115.00
MSA804142	Hydrogen Fluoride	1-50	3 (ceiling)	102.00
MSA487339	Hydrogen Sulfide	1-200	10	89.00
MSA487340	Hydrogen Sulfide	100-4000	10	89.00
MSA497663	Mercury	0.01-0.08 mg/m ³	0.025 mg/m ³	103.00
MSA813334	Methyl Ethyl Ketone	50-4000	200	115.00
MSA487341	Nitrogen Dioxide	0.5-50	3	99.00
MSA804435	Nitrogen Dioxide	2-140	3	102.00
MSA487336	Nitrous Fumes	0.5-50	N/A	99.00
MSA804425	Nitrous Fumes	2-140	N/A	99.00
MSA803946	Nitrous Fumes	10-300	N/A	102.00
MSA804426	Nitrous Fumes	50-3000	N/A	102.00
MSA497664	n-Octane	50-3000	300	99.00
MSA804140	Ozone	0.05-5	0.05 (ceiling)	99.00
MSA804429	Perchloroethylene	5-200	25	89.00
MSA487337	Perchloroethylene	10-500	25	89.00
MSA813778	Phenol	1-25	5	102.00
MSA497101	Phosphine	0.05-3	0.3	118.00
MSA485680	Phosphine	0.1-100	0.3	89.00
MSA489119	Phosphine	50-2000	0.3	89.00
MSA804418	Propane	200-4000	2500	99.00
MSA497665	Qualitest QL	Qualitative		89.00
MSA10065254	Smoke Tubes (Refill with 10 tubes)			103.00
MSA804135	Styrene	10-300	20	99.00
MSA487338	Sulfur Dioxide	0.5-25	2	99.00
MSA497662	Sulfur Dioxide	5-120	2	99.00
MSA497661	Sulfur Dioxide	100-4000	2	102.00
MSA804433	Sulfur Hexafluoride	0.5-15	1000	115.00
MSA803947	Toluene	5-1000	50	99.00
MSA487343	Trichloroethane	5-1500	350	89.00
MSA804134	Triethylamine	5-30	1	102.00
MSA803950	Vinyl Chloride	1-70	5	102.00
MSA803947	Xylene	5-2500	100	99.00



LP1200 Piston Hand Sampling Pump & Tubes



Simply select the tube for the suspected contaminant at a specified concentration range, break off the tube at both ends, insert the RAE detector tube into the piston hand pump and draw the required sample volume. The measurement can then be read directly from the scale printed on the tube.

Pump features a locking handle, graduated shaft with red line indicator for 50 ml and 100 ml sample volumes, stroke counter and simple tube break-off and disposal system. The kit version includes the pump, leather carrying case, inlet filters, spare rubber inlet and plunger gasket.

Cat. No.	Description	Rental	Purchase Price
RAE0100901000	Hand Pump	YES	\$180.00
RAE0100001000	Hand Pump Kit	YES	195.00



Total Safety hires the best safety professionals in the business to supplement your safety needs.

- Writing confined space plans
- Developing LOTO blinding plans & procedures
- Investigating all accidents & near-misses
- Permit writing
- Issuing safety plans for high-risk work
- EMT/paramedic services

DETECTOR TUBES

Cat. No.	Description	Measuring Range (ppm)	Tubes/ Box	Purchase Price
RAE1011140	Acetone	0.1-2 Vol %	10	\$34.00
RAE1010005	Ammonia	1-30	10	34.00
RAE1010010	Ammonia	5-100	10	34.00
RAE1010015	Ammonia	25-500	10	34.00
RAE1010040	Ammonia	1-15 Vol %	10	34.00
RAE1010110*	Benzene	5.0-40	5	34.00
RAE1010120	Benzene	5-100	10	34.00
RAE1010430	Carbon Dioxide	300-5,000	10	34.00
RAE1010440	Carbon Dioxide	0.05-1 Vol %	10	34.00
RAE1010445	Carbon Dioxide	0.25-3 Vol %	10	34.00
RAE1010450	Carbon Dioxide	1-20 Vol %	10	34.00
RAE1010245	Carbon Monoxide	0.2- 4 Vol %	10	34.00
RAE1010218	Carbon Monoxide	5-100	10	34.00
RAE1010230	Carbon Monoxide	20-500	10	34.00
RAE1010610	Chlorine	0.5-8	10	34.00
RAE1010620	Chlorine	5-100	10	34.00
RAE1011030	Hydrocarbons	50-1,000	10	34.00
RAE1010810	Hydrogen Chloride	1-20	10	34.00
RAE1010822	Hydrogen Chloride	20-500	10	34.00
RAE1010510	Hydrogen Fluoride	0.5-20	10	34.00
RAE1010304	Hydrogen Sulfide	0.2-3	10	34.00
RAE1010310	Hydrogen Sulfide	2.5-60	10	34.00
RAE1010315	Hydrogen Sulfide	10-120	10	34.00
RAE1010318	Hydrogen Sulfide	25-250	10	34.00
RAE1010340	Hydrogen Sulfide	0.1-2 Vol %	10	34.00
RAE1010330	Hydrogen Sulfide	100-2,000	10	34.00
RAE1010350	Hydrogen Sulfide	2-40 Vol %	10	34.00
RAE1010320	Hydrogen Sulfide	50-800	10	34.00
RAE1011320	MEK	0.02 -0.6 Vol %	10	34.00
RAE1010920*	Nitric Oxides	1-50	5	34.00
RAE1011610	Phosphine	5-50	10	34.00
RAE1011620	Phosphine	25-500	10	34.00
RAE1010715	Sulfur Dioxide	2-30	10	34.00
RAE1010730	Sulfur Dioxide	200-4,000	10	34.00
RAE1010720	Sulfur Dioxide	5-100	10	34.00
RAE1010725	Sulfur Dioxide	100-1,800	10	34.00
RAE1010740	Sulfur Dioxide	0.2-5 Vol %	10	34.00
RAE1011420	Toluene	10-300	10	34.00
RAE1012010	Water Vapor	2-10#/MMCF	10	34.00
RAE1012020	Water Vapor	6-40#/MMCF	10	34.00
RAE1011220	Xylene	10-200	10	34.00

* Includes 5 tubes, 5 pretubes and 1 connector



accuro® Bellows Hand Pump & Tubes



Manual gas detector pump used for spot measurements with Dräger tubes. Air sample is drawn through the Dräger tube using strokes. Body of the pump consists of a bellows, which is completely pushed together for the measurement. When the bellows is released, the air is drawn automatically, and the gas sample to be measured is sucked through the tube being used. The sampling process is completed when the body of the pump has opened completely. The end of a stroke is indicated by a pressure-controlled display on the body of the pump.

Contact Total Safety for additional accessories and kits.

ACCURO® BELLOWS HAND PUMP

Cat. No.	Description	Rental	Purchase Price
DRA6400000	accuro Bellows Hand Pump	YES	\$436.00

DRÄGERTUBES® SHORT-TERM DETECTOR TUBES

Cat. No.	Gas/Vapor	Measuring Range (ppm)	Qty.	Purchase Price
DRA6722101	Acetic Acid	5 – 80	10	\$79.00
DRACH22901	Acetone	100 – 12,000	10	73.00
DRA8101121	Acid Test	Qualitative	N/A	73.00
DRA6728591	Acrylonitrile	0.5 – 20	5	73.00
DRACH25301	Air Current (Smoke Tubes)	N/A	N/A	65.00
DRA8101631	Alcohol	25 – 5,000	10	90.00
DRA8101711	Ammonia	0.25 – 3	10	85.00
DRA6733231	Ammonia	2 – 30	10	74.00
DRACH20501	Ammonia	5 – 700	10	89.00
DRA6728561	Benzene	0.5 – 10	10	79.00
DRA8101841	Benzene (Specific)	0.5 – 10	5	79.00
DRA8101741	Benzene	15 – 420	10	79.00
DRA8101231	Benzene	2 – 60	10	79.00
DRA6728071	Benzene	5 – 50	10	74.00
DRACH23501	Carbon Dioxide*	0.1 – 6 Vol %	10	65.00
DRACH31401	Carbon Dioxide	0.5 – 10 Vol %	10	73.00
DRACH25101	Carbon Dioxide	1 – 20 Vol %	10	67.00
DRA8101811	Carbon Dioxide	100 – 3,000	10	69.00
DRACH20301	Carbon Dioxide	5 – 60 Vol %	10	99.00
DRA8101891	Carbon Disulfide	3 – 95	10	79.00
DRACH29901	Carbon Monoxide	0.3 – 7 Vol %	10	73.00
DRACH20601	Carbon Monoxide	10 – 3,000	10	65.00
DRACH25601	Carbon Monoxide*	5 – 700	10	69.00
DRA8101021	Carbon Tetrachloride	1 – 50	5	89.00
DRACH24301	Chlorine	0.2 – 30	10	67.00
DRA6728411	Chlorine	0.3 – 10	10	83.00
DRACH20701	Chlorine	50 – 500	10	83.00
DRA6728761	Chlorobenzene	5 – 200	5	83.00
DRA6728791	Cyanide	2 – 15 mg/m ³	10	114.00

Cat. No.	Gas/Vapor	Measuring Range (ppm)	Qty.	Purchase Price
DRA6728451	Dimethyl Sulfide	1 – 15	5	\$87.00
DRA6728381	Ethyl Benzene	30 – 600	10	79.00
DRA8101351	Ethylene Glycol	10 – 180 mg/m ³	5	87.00
DRA8101751	Formaldehyde	2 – 40	10	105.00
DRACH31801	Hydrazine	0.1 – 10	10	87.00
DRA8103571	Hydrocarbons (Propane)	0.1 – 1.3 Vol %	10	73.00
DRA8103581	Hydrocarbons	2 – 24 mg/l	10	79.00
DRA8103481	Hydrochloric Acid	0.2 – 20	10	99.00
DRACH29501	Hydrochloric Acid	1 – 10	10	88.00
DRACH25701	Hydrocyanic Acid	2 – 150	10	74.00
DRACH30301	Hydrogen Fluoride	1.5 – 15	10	87.00
DRACH28201	Hydrogen Sulfide and Sulfur Dioxide	0.02 – 7 Vol %	10	114.00
DRACH28101	Hydrogen Sulfide	0.2 – 7 Vol %	10	88.00
DRA8101991	Hydrogen Sulfide	0.2 – 6	10	79.00
DRA8101831	Hydrogen Sulfide*	1 – 200	10	65.00
DRA6728821	Hydrogen Sulfide	2 – 200	10	73.00
DRACH29801	Hydrogen Sulfide	5 – 600	10	65.00
DRA6728981	Mercaptan	0.5 – 5	10	101.00
DRACH23101	Mercury	0.05 – 2 mg/m ³	10	79.00
DRA6728161	Ethyl Acrylate	5 – 200	10	99.00
DRA8101671	Methyl Bromide	0.5 – 30	10	140.00
DRA6728211	Methyl Bromide	3 – 100	10	79.00
DRACH27301	Methyl Bromide	5 – 50	10	99.00
DRA6723301	Monostyrene	10 – 200	10	103.00
DRACH20001	Natural Gas Test (Methane)	Qualitative	5	67.00
DRA6728311	Nitric Acid	1 – 50	10	88.00
DRACH30001	Nitrogen Dioxide*	0.5 – 25	10	68.00
DRACH29401	Nitrous Gases	0.5 – 10	10	65.00
DRACH31001	Nitrous Gases	2 – 100	10	67.00
DRACH25903	Organic Basic Nitrogen Compounds	1 mg/m ³	10	105.00
DRA8103261	Oxygen	5 – 23 Vol %	8	166.25
DRA6733181	Ozone	0.05 – 1.4	10	79.00
DRACH21001	Ozone	10 – 300	9	99.00
DRA8101501	Perchloroethylene	2 – 300	10	73.00
DRA8101691	Petroleum Hydrocarbons	10 – 300	10	99.00
DRA6730201	Petroleum Hydrocarbons	100 – 2,500	10	73.00
DRA8101641	Phenol	1 – 20	10	89.00
DRA8101521	Phosgene	0.02 – 1	10	99.00
DRA8101611	Phosphine	0.1 – 1.0	10	99.00
DRA8103341	Phosphine in acetylene	0.1 – 15	10	105.00
DRACH21201	Phosphine	15 – 1,000	10	79.00
DRA6728461	Phosphoric Acid Esters	0.05	10	130.00
DRACH25301	Smoke Tubes	N/A	N/A	61.00
DRA6727101	Sulfur Dioxide	0.1 – 3	10	79.00
DRA6728491	Sulfur Dioxide	0.5 – 25	10	73.00
DRACH24201	Sulfur Dioxide	20 – 2,000	10	79.00
DRA8101531	Sulfur Dioxide	50 – 8,000	10	79.00
DRA6728781	Sulfuric Acid	1 – 5 mg/m ³	9	111.00
DRA8101661	Toluene	5 – 300	10	89.00
DRA8101701	Toluene	50 – 400	10	69.00
DRACH21101	Trichloroethane	50 – 600	5	67.00
DRA6728541	Trichloroethylene	2 – 250	10	73.00
DRA8101881	Trichloroethylene	50 – 2,000	10	73.00
DRA8101721	Vinyl Chloride	0.5 – 30	5	79.00
DRACH19601	Vinyl Chloride	100 – 3,000	10	99.00
DRA6733161	Xylene	10 – 400	10	73.00

* Indicates SEI Certification

SENSIDYNE
Industrial Health & Safety Instrumentation

Model AP-20S Gas Detector Tube Pump Kit

Highly portable and rugged pump that gives the user the ability to perform spot test for hundreds of different gases and chemicals at very low cost per sample. No need for calibration or electrical power; pump minimizes the possibility of user error. Kit includes a pump, carrying bag, spare tube holders, lubricant and an instruction manual.



Cat. No.	Description	Rental	Purchase Price
SEN83000101	AP-20S Tube Pump Kit	YES	\$395.00

SENSIDYNE
Industrial Health & Safety Instrumentation

Deluxe HazMat-III Kit

Portable hazardous material detection kit provides on-the-spot detection of numerous airborne contaminants. The lightweight and rugged kit utilizes a 2-step approach for rapid identification with a primary test that uses 2 multi-stage qualitative tubes and a special color chart to determine nearly 70 compounds in as little as 2 – 3 minutes. Kit requires no electrical power or user calibration and includes a hard-shell, corrosion-resistant polyethylene case.



Cat. No.	Description	Purchase Price
SEN7013627	Deluxe HazMat-III Kit	\$1,919.50

DETECTOR TUBES

Cat. No.	Gas/Vapor	Measuring Range (ppm)	Qty.	Purchase Price
SEN133SB	Acetaldehyde	5 – 140	10	\$66.25
SEN216S	Acetic Acid	1 – 150	10	66.25
SEN102SD	Acetone	40 – 5,000	10	66.25
SEN280S	Acetylene and Ethylene	20 – 20,000 (Acetylene) 200 – 2,000 (Ethylene)	2 x 5	66.25
SEN128SD	Acrylonitrile	0.25 – 20	2 x 5	66.25
SEN105SC	Ammonia	5 – 260	10	66.25
SEN105SD	Ammonia	0.2 – 20	10	66.25
SEN118SC	Benzene	1 – 100	10	66.25
SEN118SB	Benzene (in presence of Hydrocarbons)	5 – 200	2 x 5	66.25
SEN168SC	Butadiene	2.5 – 100	10	66.25
SEN126SA	Carbon Dioxide	0.1 – 5.2 Vol %	10	66.25
SEN126SF	Carbon Dioxide	100 – 4,000	10	66.25
SEN106SA	Carbon Monoxide	5 – 1,000	10	66.25
SEN106SC	Carbon Monoxide	1 – 50	10	66.25
SEN147S	Carbon Tetrachloride	0.5 – 60	2 x 5	66.25
SEN239S	Carbonyl Sulfide	5 – 60	2 x 5	66.25
SEN109SB	Chlorine	0.1 – 10	10	66.25
SEN152S	Chloroform	23 – 500	10	66.25
SEN180S	Dichloromethane (Methylene Chloride)	10 – 1,000	2 x 5	66.25
SEN227S	Dimethyl, Ethyl and Methyl Amine	1 – 20	10	66.25
SEN190U	Ethyl and Methyl Cellosolve	5 – 500	10	66.25
SEN104SA	Ethyl Alcohol	0.05 – 5 Vol %	10	66.25
SEN179S	Ethyl Benzene	10 – 500	10	66.25
SEN122SC	Ethylene Oxide	1 – 15	2 x 5	66.25
SEN171SB	Formaldehyde	1 – 35	2 x 5	66.25
SEN171SC	Formaldehyde	0.1 – 4	10	66.25
SEN110S	Gasoline	0.05 – 0.6 Vol %	10	66.25
SEN219S	Hydrazine	0.1 – 10	10	66.25
SEN187S	Hydrocarbons	50 – 1,400	10	66.25
SEN173SB	Hydrogen Chloride	0.4 – 40	10	66.25
SEN112SB	Hydrogen Cyanide	0.5 – 100	10	66.25

Cat. No.	Gas/Vapor	Measuring Range (ppm)	Qty.	Purchase Price
SEN156S	Hydrogen Fluoride	1 – 30	10	\$66.25
SEN120SD	Hydrogen Sulfide	1 – 60	10	66.25
SEN120SF	Hydrogen Sulfide	25 – 1,000	10	66.25
SEN120U	Hydrogen Sulfide	0.2 – 6.0	10	66.25
SEN131	Inorganic Gases	Qualitative	10	66.25
SEN150U	Isopropyl Alcohol	20 – 1,200	10	66.25
SEN130U	Mercaptans	0.5 – 10	10	66.25
SEN142S	Mercury Vapor	0.1 – 10 mg/m ³	10	66.25
SEN119U	Methyl Alcohol	20 – 1,000	10	66.25
SEN157SC	Methyl Bromide	0.5 – 10	1 x 5	66.25
SEN160S	Methyl Chloroform (1, 1, 1-Trichloroethane)	15 – 4000	2 x 5	66.25
SEN139U	Methyl Ethyl Ketone	20 – 1,500	10	66.25
SEN155U	Methyl Isobutyl Ketone	5 – 300	10	66.25
SEN233S	Nitric Acid Vapor	1 – 20	10	66.25
SEN174A	Nitric Oxide and Nitrogen Dioxide	10 – 30 (Nitric Oxide) 1 – 40 (Nitrogen Dioxide)	5	66.25
SEN117SC	Nitrogen Dioxide	0.1 – 10	10	66.25
SEN175U	Nitrogen Oxides	0.5 – 30	10	66.25
SEN186B	Organic Vapors	Qualitative	10	66.25
SEN159SB	Oxygen	1.5 – 24 Vol %	2 x 5	66.25
SEN182U	Ozone	0.05 – 3	10	66.25
SEN146S	Phosgene	0.1 – 20	10	66.25
SEN121U	Phosphine and Arsine	0.5 – 2	10	66.25
SEN158S	Styrene	2.5 – 300	10	66.25
SEN103SD	Sulfur Dioxide	1 – 60	10	66.25
SEN103SE	Sulfur Dioxide	0.25 – 10	10	66.25
SEN135SB	Tetrachloroethane	1 – 10	10	66.25
SEN135SA	Tetrachloroethane (Perchloroethane)	5 – 300	10	66.25
SEN124SA	Toluene	10 – 500	10	66.25
SEN134SA	Trichloroethane	5 – 300	10	66.25
SEN134SB	Trichloroethane	1 – 16	10	66.25
SEN132SC	Vinyl Chloride	0.1 – 12	2 x 5	66.25
SEN177SA	Water Vapor	1.7 – 33.8 mg/l	10	66.25
SEN143SB	Xylene	5 – 200	10	66.25

**ALTAIR® QuickCheck™ Station**

Designed to test ALTAIR® and ALTAIR® Pro Single Gas Detectors (see page 96), verifying instrument gas response (bump check) and alarm functionality by checking the instrument's visual, audible and vibrating alarms. Performs visual and vibrating alarm activation, audible alarm activation and verification, sensor response to gas and general pass/fail indication.

Station is for use with all ALTAIR Single Gas Detector versions and the following ALTAIR Pro Single Gas Detectors: O₂, H₂S, CO, CO Fire, CO Steel, SO₂, NO₂, Cl₂, ClO₂,

NH₃ and HCN. After testing, the station will record the date, time and result of the bump test internally. Able to store up to 25 events in the ALTAIR and up to 50 events in the ALTAIR Pro Detectors.

Available in automatic and manual versions. Automatic version uses an automatic gas delivery system and comes complete with regulator and tubing. Automatic version features a 0.25 LPM automatic system that pulls gas as it is needed. Manual version uses a manual regulator and requires regulator and tubing, which is purchased separately. Manual version features a 0.25 LPM manual regulator that user turns on during testing. Comes standard with AC power supply. Vehicle power supply sold separately.

Cat. No.	Description	Purchase Price
MSA10076710	HCN, Automatic	\$594.00
MSA10076707	NH ₃ , Automatic	594.00
MSA10076713	Cl ₂ , ClO ₂ , Automatic	594.00
MSA10076704	O ₂ , CO, H ₂ S, SO ₂ , NO ₂ , Automatic	569.00
MSA10076698	HCN, Manual	272.00
MSA10076695	NH ₃ , Manual	272.00
MSA10076701	Cl ₂ , ClO ₂ , Manual	272.00
MSA10076692	O ₂ , CO, H ₂ S, SO ₂ , NO ₂ , Manual	255.00

ACCESSORIES

Cat. No.	Description	Purchase Price
MSA10077384	Regulator Tubing	\$14.00
MSA10075893	Automatic Gas Regulator	180.00
MSA467895	Manual Regulator	207.00
MSA10049410	Vehicle Power Supply	128.00

**Galaxy® Automated Test System**

Compatible with MSA's Sirius® (see page 89) multigas detectors and the ALTAIR® 4/4X and ALTAIR® 5/5X (see page 88) multigas detectors.

Provides automated record-keeping and calibration for MSA portable gas detection instruments. System is housed in a high-strength polycarbonate plastic and is stand-alone, requiring no computer, controller or network interface. Optional cylinder holder allows for simple connection of the gas cylinder to the Galaxy® Test Stand, completing the set-up. Up to 10 test modules can be interconnected. Each test stand has automated record keeping capabilities. Where applicable, instrument charging is available and automatic.

An industry-standard memory flash card is available for data retention of more than 1000 individual test records, and an optional printer is also available. A network interface is available that will provide calibration records, due dates for instrument calibration and e-mail notifications. Alkaline battery pack optional. All-inclusive, 2-year warranty.

Standard accessories include power supply instruction manual.

Standard Stand-Alone Kits include standard accessories plus regulator.

Smart Stand-Alone Kits include standard accessories plus regulator and memory card.

Portable Kits include regulator and battery pack.

Network Kits include regulator and wired Ethernet access.

Additional Galaxy® kits and other accessories available. Contact Total Safety for more information.

SIRIUS® GALAXY® KITS

Cat. No.	Description	Rental	Purchase Price
MSA10061050	Basic System		\$1,425.00
MSA10061810	Standard System	YES	1,925.00
MSA10061811	Standard System and Charging		2,075.00
MSA10061814	Smart System		1,975.00
MSA10061818	Portable System		2,075.00
MSA10061820	Wired Network System		2,525.00

ALTAIR® AND ALTAIR PRO® GALAXY® KITS

Cat. No.	Description	Purchase Price
MSA10078252	Basic System	\$1,425.00
MSA10078253	Standard System	1,675.00
MSA10078255	Smart System	1,725.00
MSA10078257	Portable System	1,825.00
MSA10078259	Wired Network System	2,275.00



Calibration Check Kits

When used with the appropriate calibration check gas, these check kits offer a quick, convenient and economical method of checking the response of MSA gas detection instruments.

The **Model R and RP Check Kits** use compressed calibration check gas for combustible gas detectors and certain multigas detectors. Kits include a regulating valve, tubing, sensor adapters as needed, and a carrying case. Depending on the kit, the regulator supplies gas at either 0.25 or 1.5 LPM. Cylinders for use with the R models are small, steel containers containing 20 L of gas mixture. RP cylinders are steel and contain 100 L of gas mixture. Note: hydrogen sulfide cylinders contain 58 L.

Aerosol-Type Kits, also for combustible gas, use a convenient 3-L pressurized can for added portability. Kit includes a 3-L can of 3% methane in air (although other gases are available), 3 rubber bladders, a bladder coupler, tubing, tube adapter and gasket and pinch clamp.

For Use with Instrument	Cat. No.	Cal. Kit Required	Price	Cat. No.	Cal. Gas Cylinder	Price
Gascope 62	MSA476609	Model R	\$331.00	MSA459945	Methane 40% LEL	\$70.00
				MSA459942	Methane 50% LEL	70.00
Explosimeters, (all) Model 2A	MSA477150	Model RP	319.00	MSA491041	Methane 50% LEL	207.00
	MSA454380	Aerosol	213.00	MSA96329	Methane 40% LEL	69.00
Mini CO, CO Indicator, Model IV				MSA477888	CO 20 ppm	70.00
				MSA459944	CO 150 ppm	70.00
				MSA461769	CO 300 ppm	70.00
Mini H ₂ S, H ₂ S Indicator	MSA477149	Model RP	291.00	MSA467898	H ₂ S 10 ppm	207.00
				MSA467897	H ₂ S 40 ppm	207.00
Combustible Gas & Oxygen Alarms	MSA476609	Model R	295.00	MSA476304	Pentane 50% LEL, Oxygen 15%	70.00
	MSA477150	Model RP	331.00	MSA478192	Pentane 60% LEL, Oxygen 15%	207.00
MicroGard with Pump Module	MSA476609	Model R	331.00	MSA476304	Pentane 50% LEL, Oxygen 15%	70.00
	MSA477150	Model RP	319.00	MSA459942	Methane 50% LEL	70.00
MicroGard, Diffusion Mode	MSA477149	Model RP	291.00	MSA478192	Pentane 60% LEL, Oxygen 15%	207.00
				MSA476304	Pentane 50% LEL, Oxygen 15%	70.00
				MSA459942	Methane 50% LEL	70.00
Passport LEL, O ₂ & CO Monitor	MSA477149	Model RP	291.00	MSA478191	Pentane 60% LEL, Oxygen 15%, CO 60 ppm	207.00
Passport LEL & Oxygen Monitor	MSA477149	Model RP	291.00	MSA478192	Pentane 60% LEL, Oxygen 15%	207.00
Passport H ₂ S Monitor	MSA477149	Model RP	291.00	MSA467898	H ₂ S 10 ppm	207.00



Squirt™ Gas Bump Tester

The Squirt™ Gas Bump Tester uses a lightweight disposable cylinder to provide an instant calibration check to ensure your instrument is working properly. Simply release a small shot of gas into the assembly and a reading is provided in just seconds. Kit includes assembly, 3 gas balloons and carrying case. Gas cylinders must be ordered separately.

Cat. No.	Description	Purchase Price
MSA813411	Bump Test Kit	\$80.00
MSA814350	Gasport Cylinder (CO ₂ , O ₂ , CH ₄)	70.00
MSA814497	Passport Cylinder (CO ₂ , O ₂ , CH ₄)	70.00





MicroDock II Docking Station



Self-contained, fully automatic calibration and bump testing station provides simultaneous management of up to 10 modules. Automatically verifies performance of audible and visual alarms without the use of a computer. Stores and updates calibration records in the station and datalogging

instrument. Operates via a 6V wall adapter or alkaline batteries. Features multi-language support. Requires a demand flow regulator.

Basic System includes 110-240V AC power adaptor, installed memory card, software, USB cable, fresh air inlet filter, alkaline batteries, calibration gas hoses and 2 gas cylinder inlet fittings.

Portable System Kits include the above basic system contents plus one 34-L calibration gas cylinder, a demand flow regulator and a rugged carrying case.

2-year warranty.

Complies with the FCC Part 15 and ICES-003 Canadian EMI requirements.

BASIC SYSTEM

Cat. No.	Description	Rental	Purchase Price
BWTDCK221E00G	GasAlertClip Extreme H ₂ S	YES	\$1,495.00
BWTDCK221F00G	GasAlertClip Extreme SO ₂	YES	1,495.00
BWTDCK221G00G	GasAlertClip Extreme CO		1,495.00
BWTDCK221H00G	GasAlertClip Extreme O ₂		1,495.00
BWTDCK221C1K00G	GasAlertMicro5, GasAlert-Micro5 PID and GasAlert-Micro5 IR	YES	1,695.00

PORTABLE SYSTEM KITS

Cat. No.	Description	Purchase Price
BWTDCK221E01G	GasAlertClip Extreme H ₂ S	\$1,925.00
BWTDCK221F01G	GasAlertClip Extreme SO ₂	1,925.00
BWTDCK221G01G	GasAlertClip Extreme CO	1,925.00
BWTDCK221H01G	GasAlertClip Extreme O ₂	1,925.00
BWTDCK221C1K01G	GasAlertMicro5, GasAlertMicro5 PID and GasAlertMicro5 IR	2,125.00

ACCESSORIES

Cat. No.	Description	Rental	Purchase Price
BWTREGDF1	Demand Flow Regulator*	YES	\$325.00
BWTWMADOCK	Calibration Gas Wall Mount		22.00

*Each cylinder of gas connected to the system requires an individual demand flow regulator – sold separately.



DS2 Docking Station™

- For Up to 100 ISC Satellite Gas Monitors
- Requires Docking Server



Ideal for automatic instrument calibration, record-keeping, diagnostics and recharging. Minimizes time required to manage post-purchase responsibilities and reduces liability. Features two-way communication with a Docking Station Server (not included) that manages up to 100 instrument docking satellites (IDSs) through Ethernet or wireless connections, and a graphical user interface to monitor facility-wide network. Free-standing, self-contained IDS has a multi-lingual LCD display, status LEDs and an audible alarm that allows plant-wide distribution for autonomous operation or up to 5 IDSs can be grouped together to share calibration gas cylinders. Stores instrument data in a central database.

Cat. No.	Description	Rental	Purchase Price
ISC18106724ABC	MX6 iBrid Instrument Docking Station	YES	Call for Quote
ISC18108630ABC	Ventis MX4 DS2	YES	Call for Quote
ISC18105551ABC	iTX Instrument Docking Station	YES	Call for Quote
ISC18106302ABC	GasBadge Pro Docking Station		Call for Quote
ISC18107698ABC	GasBadge Plus Docking Station		Call for Quote

*All ABC extensions require:

A = Wireless Option

B = Number of iGas Readers

C = Power Cord Option

**INDUSTRIAL
SCIENTIFIC**

V•CAL™ Calibration Station

Capable of calibrating and function (bump) testing the Ventis MX4 instrument (see page 93). Unit is transportable and provides stand-alone operation with no PC required. When used in conjunction with a PC, the station automatically downloads calibration and bump test reports via a standard USB connection (USB 2.0 recommended). Users can access alarm event data to keep track of gas exposure or to monitor high-risk areas. A PC connection also makes it easy to configure the Ventis MX4's preferences. An optional serial data dot matrix printer automatically prints calibration and bump test reports as proof of regulatory compliance with records of instrument serial numbers, pass/fail indications, bump test readings and full span readings for the Ventis MX4. Calibration gas and regulators sold separately.



Cat. No.	Description	Rental	Purchase Price
ISC18108631AB*	V-Cal Calibration Station	YES	Call for Quote
ISC18107664ABC**	V-Cal Six-Unit Calibration Station	YES	Call for Quote
ISC18107763	Serial Data Dot Matrix Printer for the V-Cal – 120 V AC powered		\$595.00
ISC17124348	Wall/Desk Mount Cylinder Holder		54.00
ISC18102509	Demand Flow Regulator for 34L/58L/103L Aluminum Cylinders		420.00
ISC18103556	Demand Flow Regulator for 650L Cylinder		420.00

Cat. No.	Description	Purchase Price
ISC18103937	34L (100 ppm CO, 25 ppm H ₂ S, 25% LEL pentane, 19% O ₂)	\$220.00
ISC18102187	58L (100 ppm CO, 25 ppm H ₂ S, 25% LEL pentane, 19% O ₂)	278.30
ISC18103366	650L (100 ppm CO, 25 ppm H ₂ S, 25% LEL pentane, 19% O ₂)	1,705.00
ISC18105536	34L (100 ppm CO, 25 ppm H ₂ S, 50% LEL methane, 19% O ₂)	220.00
ISC18102242	58L (100 ppm CO, 25 ppm H ₂ S, 2.5% LEL methane, 19% O ₂)	278.30
ISC18106799	58L (25 ppm H ₂ S, 5 ppm SO ₂ , 25% LEL pentane, 19% O ₂)	324.50
ISC18106807	58L (25 ppm H ₂ S, 5 ppm SO ₂ , 25% LEL methane, 19% O ₂)	278.30
ISC18106773	58L (100 ppm CO, 5 ppm NO ₂ , 25% LEL pentane, 19% O ₂)	278.30
ISC18106781	58L (100 ppm CO, 5 ppm NO ₂ , 25% LEL methane, 19% O ₂)	278.30

*A = Instrument Type: 0 = Ventis,
1 = Ventis with Pump
B = Power Cord: 0 = US, 1 = UK, 2 = EU,
3 = AUS, 4 = ITA, 5 = DEN, 6 = SWZ

**AB = Number of Ventis (A) and Ventis with Pump (B) Instruments
06 = 0 Ventis and 6 Ventis with Pump
60 = 6 Ventis and 0 Ventis with Pump
C = Power Cord: 0 = Universal with US, UK, EU, AUS Plug Adapters

**INDUSTRIAL
SCIENTIFIC**

Cal Plus™ Calibration System

- Standalone Portable System for GasBadge® Plus



Simple, 2-button operation allows the user to quickly and easily calibrate or function (bump) test the GasBadge® Plus monitor. The large, multilingual LCD display and the LED indicators show if the instrument passed or failed the desired function.

Standalone system. PC not required to operate. Generates calibration and bump test reports automatically with a standard USB connection. The unit also downloads alarm events and instrument details allowing the user to quickly and easily configure instrument preferences.

Available with built-in dot-matrix printer that automatically prints calibration and bump test reports. The reports provide permanent documentation of instrument serial numbers, pass/fail indications, bump test readings and full span readings. Cal Plus™ software for PCs included.

Wall-mountable. Calibration gas and regulator not included.

Cat. No.	Description	Rental	Purchase Price
ISC1810634400	Calibration Station		\$980.00
ISC1810634410	Calibration Station w/ On-Board Printer		1,350.00
ISC17117722	Serial Data Dot Matrix Printer, 120V AC Powered		527.90
ISC18102163	Calibration Gas – CO, 100 ppm, 103 L		148.50
ISC18100859	Calibration Gas – H ₂ S, 25 ppm, 58 L		173.80
ISC18102219	Calibration Gas – NO ₂ , 5 ppm, 58 L		418.00
ISC18102222	Calibration Gas – SO ₂ , 5 ppm, 58 L		173.80
ISC17124348	Wall Mounted Gas Cylinder Holder		54.00
ISC18102509	Demand Flow Regulator	YES	420.00



INDUSTRIAL SCIENTIFIC

Calibration Equipment

Industrial Scientific Calibration Kits come equipped with everything necessary to keep instruments operating accurately and reliably. Kits contain NIST-traceable span gases. Regulators provide proper flow rate for calibrating instruments. Always use the appropriate regulator for the application as recommended.

Cat. No.	Description	Purchase Price
ISC18105833	552 L Demand Flow Regulator, 590 CGA with iGas Pressure Switch	\$500.00
ISC18105858	650 L Demand Flow Regulator, 330 CGA with iGas Pressure Switch	500.00
ISC18106740	Demand Flow Regulator, 660 CGA with iGas Pressure Switch	500.00
ISC18106757	Demand Flow Regulator, 705 CGA with iGas Pressure Switch	720.00

MISCELLANEOUS EQUIPMENT

Cat. No.	Description	Purchase Price
ISC17041807	Calibration Log (Tablet of 50 Sheets)	\$4.00
ISC17050734	Calibration Log, TMX, LTX, STX (Tablet of 50 Sheets)	6.00
ISC17045873	Calibration Label	1.00
ISC17050831	Calibration Cup, 300 Series	6.00
ISC17042839	Calibration Cup, 400 Series	6.00
ISC17059189	Calibration Cup, T82/T80 (Not Included with Instrument)	46.00
ISC17128489	Calibration Cup, MX6 iBrid	1.00
ISC17092339	Calibration Cup, iTX	7.00
ISC17108622	Calibration Cup, M40	3.00
ISC17124033	Calibration Cup, GasBadge Plus/Pro	27.00
ISC17056326	Bump Cylinder Adapter for CO Breath Sampler	88.50
ISC17054578	Calibration Coupling for MDU420, CDU440	105.00
ISC17037961	Carrying Case for 2 Cylinders (58/103 L)	75.90
ISC18100149	Carrying Case for 2 Cylinders (34 L) with 0.5 LPM Regulator	181.50
ISC17124348	Wall-Mount Gas Cylinder Holder	54.00
ISC17113275	Cylinder Recycling Tool (58 L, 103 L)	134.75
ISC17113283	Cylinder Recycling Tool (34 L)	134.75



Make one call to **Total Safety** for your calibration gas sales. Get all the consumables you need to keep your gas detection instruments running with assured quality and traceability.



AutoRAE Lite Station

- For the ToxiRAE 3 and QRAE II



ToxiRAE 3 Model



QRAE II Model

High-speed and low-operation-cost bump test and calibration station for either the QRAE II or the ToxiRAE 3 that stores data in a text file that can be opened by any spreadsheet or editor. Station operates on a low-pressure system with built-in pump. Bump tests can be performed in 12 sec and full calibration in 2 min.

Station features a molded rubber insert, a "no gas flow" alarm, a demand-flow system to prevent accidental leaks and optional battery operation. Provides full documentation storage with tamper-proof data files.

CE certified.

Cat. No.	Description	Rental	Purchase Price
RAET010100000	For ToxiRAE 3	YES	\$1,231.00
RAET010001000	For QRAE II		1,334.00



OSHA – 1910.146 Permit-required Confined Spaces

Test Before Entering with a Gas Detector

Many workplaces contain spaces that are considered “confined” because their configurations hinder the activities of employees who must enter, work in and exit them. A confined space has limited or restricted means for entry or exit, and it is not designed for continuous employee occupancy.

Confined spaces include, but are not limited to:

- Underground vaults
- Tanks
- Storage bins
- Manholes
- Pits
- Silos
- Process vessels
- Pipelines

OSHA uses the term “permit-required confined space” (permit space) to describe a confined space that has one or more of the following characteristics:

- Contains or has the potential to contain a hazardous atmosphere
- Contains a material that has the potential to engulf an entrant
- Has walls that converge inward or floors that slope downward and taper into a smaller area which could trap or asphyxiate an entrant
- Contains any other recognized safety or health hazard, such as unguarded machinery, exposed live wires or heat stress

Atmospheric testing is required for two distinct purposes:

- Evaluation of the hazards of the permit space
- Verification that acceptable entry conditions for entry into that space exist



Honeywell IQ Express™ Docking Station

- For Honeywell Gas Monitors



Fully automated bump test and calibration station that performs up to 8 critical tests including: gas sensor identification, instrument performance and record-keeping procedures in under 20 sec. If the bump test fails, station will automatically perform a full calibration without intervention from the user. In addition to the tests performed, station will also automatically perform instrument identification, battery test and audible alarm test (if applicable).

Station can be connected to the PC via USB to download instrument data. Connect up to 4 stations to a single gas supply.

IQ EXPRESS FOR MULTIPRO™

Cat. No.	Description	Purchase Price
SPR54499000	Standard Version	\$895.00
SPR54499000N	Ethernet-Ready Version	1,045.00



PORTAGAS Calibration Gas

PortaGas has replaced traditional disposable cylinders with PortaGreen refillable/recyclable cylinders that require no hole drilling or valve crushing before disposal. There is no return fee for the Strip & ShipSM program. Just make sure the cylinder is empty, strip off the shrink wrap and return in the provided shipping tube.

All gases are made to the highest NIST traceable standards and have the longest industry shelf lives. Certified calibration gases are available in the new PortaGreen 34-, 58-, 100-, and 116-L cylinders. The standard 17- and 34-L steel cylinders are available in the Non-PortaGreen package. Approximate concentrations are listed below. MSDS available upon request.



17L & 34L STANDARD STEEL CYLINDERS*

Description	17-L Cylinders		34-L Cylinders	
	Cat. No.	Price	Cat. No.	Price
Air Zero THC 1 ppm	17PU-09000	\$37.00	34PU-09000	\$54.00
Benzene 5 ppm, Balance Air	17SG-00127	41.00	34SG-00127	58.00
Carbon Dioxide 2000 ppm, Balance Air	17SG-00188	41.00	34SG-00188	58.00
Carbon Monoxide 200 ppm, Balance Air	17SG-00101	41.00	34SG-00101	58.00
Carbon Monoxide 60 ppm, Balance Air	17SG-00102	41.00	34SG-00102	58.00
Cocktail CO ₂ (35% by Vol.), Methane (50% by Vol.), Balance Nitrogen	17DU-02008	41.00	34DU-02008	58.00
Hexane 0.30% by Vol. (25% LEL), Balance Air	17SG-00104	41.00		
Hexane 0.48% by Vol. (40% LEL), Balance Air	17SG-00190	41.00	34SG-00190	58.00
Hexane 100 ppm, Balance Air	17SG-00106	41.00	34SG-00106	58.00
Hydrogen 0.1% by Vol. (1000 ppm), Balance Air	17SG-00109	41.00	34SG-00109	58.00
Hydrogen 0.2% by Vol. (50% LEL), Balance Air	17SG-00224	41.00	34SG-00224	58.00
Isobutylene 100 ppm, Balance Air	17SG-00112	41.00	34SG-00112	58.00
Methane 100 ppm, Balance Air	17SG-00117	41.00	34SG-00117	58.00
Methane 500 ppm, Balance Air	17SG-00118	41.00	34SG-00118	58.00
Methane 0.1% by Vol. (1000 ppm), Balance Air	17SG-00128	41.00	34SG-00128	58.00
Methane 1% by Vol., Balance Air	17SG-00119	41.00	34SG-00119	58.00
Methane 1.25% by Vol. (25% LEL), Balance Air	17SG-00120	41.00	34SG-00120	58.00
Methane 2.5% by Vol. (50% LEL), Balance Air	17SG-00121	41.00	34SG-00121	58.00
Nitrogen, Pure 99.8% by Vol	17PU-09007	41.00	34PU-09007	58.00
Oxygen 500 ppm, Balance Nitrogen	17SG-00270	41.00	34SG-00270	58.00
Oxygen 4.0% by Vol., Balance Nitrogen	17SG-00170	41.00	34SG-00170	58.00
Pentane 0.75% by Vol., Balance Air	17SG-00124	41.00	34SG-00124	58.00
Pentane 0.75% by Vol., O ₂ (15% by Vol.), Balance Nitrogen	17DU-02002	41.00	34DU-02002	58.00
Propane 50% by Vol., Balance Nitrogen	17SG-02036	41.00	34SG-02036	58.00

34L & 58L PORTAGREEN CYLINDERS*

Description	34-L Cylinders		58-L Cylinders	
	Cat. No.	Price	Cat. No.	Price
Ammonia 25 ppm, Balance Air	34SG-00242	\$284.00	58SG-00130	\$374.00
Benzene 5 ppm, Balance Air			58SG-00127	102.00
Chlorine 10 ppm, Balance Nitrogen	34SG-00131	268.00	58SG-00131	369.00
Cocktail H ₂ S (10 ppm), CO (300 ppm), CH ₄ (1.45% by Vol.-29% LEL), O ₂ (15% by Vol.), Balance Nitrogen	34SG-04002	223.00	58QD-04002	274.00
Cocktail H ₂ S (10 ppm), CO (300 ppm), Pentane Simulate (58% LEL), O ₂ (15% by Vol.), Balance Nitrogen	34QD-04006	223.00	58QD-04006	274.00
Cocktail H ₂ S (20 ppm), CO (60 ppm), CH ₄ (1.45% by Vol.-29% LEL), O ₂ (15% by Vol.), Balance Nitrogen	34QD-04034	223.00	58QD-04034	274.00
Cocktail H ₂ S (20 ppm), CO (60 ppm), Pentane Simulate (58% LEL), O ₂ (15% by Vol.), Balance Nitrogen	34QD-04010	223.00	58QD-04010	274.00
Cocktail H ₂ S (25 ppm), CO (50 ppm), CH ₄ (1.62% by Vol.-32.4% LEL), O ₂ (18% by Vol.), Balance Nitrogen	34QD-04016	223.00	58QD-04016	274.00
Cocktail H ₂ S (25 ppm), CO (50 ppm), CH ₄ (2.5% by Vol.-50% LEL), O ₂ (18% by Vol.), Balance Nitrogen			58QD-04017	274.00
Cocktail H ₂ S (25 ppm), CO (50 ppm), CH ₄ (2.5% by Vol.-50% LEL), O ₂ (20.9% by Vol.), Balance Nitrogen	34QD-04023	223.00	58QD-04055	274.00
Cocktail H ₂ S (25 ppm), CO (50 ppm), Pentane Simulate (50% LEL), O ₂ (18% by Vol.), Balance Nitrogen	34QD-04056	223.00	58QD-04056	274.00
Cocktail H ₂ S (25 ppm), CO (50 ppm), Propane Equivalent (50% LEL), O ₂ (18% by Vol.), Balance Nitrogen	34QD-04004	223.00	58QD-04004	274.00
Cocktail H ₂ S (25 ppm), CO (100 ppm), Pentane (25% LEL), O ₂ (19% by Vol.), Balance Nitrogen	34QD-04014	223.00	58QD-04014	274.00
Cocktail H ₂ S (25 ppm), CO (100 ppm), CH ₄ (2.5% by Vol.-50% LEL), O ₂ (18% by Vol.), Balance Nitrogen			58QD-04015	274.00
Cocktail CO ₂ (35% by Vol.), Methane (50% by Vol.), Balance Nitrogen			58DU-02008	102.00
Hexane 100 ppm, Balance Air			58SG-00106	102.00
Hydrogen 2% by Vol. (50% LEL), Balance Air			58SG-00224	102.00
Hydrogen Cyanide, 10 ppm, Balance Nitrogen	34SG-00132	268.00	58SG-00132	330.00
Hydrogen Sulfide 6 ppm, Balance Nitrogen Balance			58SG-00342	151.00
Hydrogen Sulfide 10 ppm, Balance Nitrogen Balance	34SG-00133	120.00	58SG-00133	151.00
Hydrogen Sulfide 25 ppm, Balance Nitrogen	34SG-00165	120.00	58SG-00134	151.00
Hydrogen Sulfide 25 ppm, Balance Air			58SG-00164	151.00
Hydrogen Sulfide 50 ppm, Balance Air			58SG-00173	151.00
Isobutylene 100 ppm, Balance Air			58SG-00112	102.00
Methane 100 ppm, Balance Air			58SG-00117	102.00
Methane 500 ppm, Balance Air			58SG-00118	102.00
Methane 2.5% by Vol. (50% LEL), Balance Air			58SG-00121	102.00
Nitric Oxide 25 ppm, Balance Nitrogen Balance	34SG-00243	268.00	58SG-00243	330.00
Nitrogen, Pure 99.8% by Vol.			58SG-09007	98.00
Nitrogen Dioxide 10 ppm, Balance Nitrogen	34SG-00214	268.00	58SG-00214	330.00
Nitrogen Dioxide 25 ppm, Balance Nitrogen	34SG-00243	268.00	58SG-00243	330.00
Propane 50% by Vol., Balance Nitrogen			58SG-02036	102.00
Sulfur Dioxide 10 ppm, Balance Nitrogen	34SG-00138	120.00	58SG-00138	151.00

*Additional cylinder sizes and gases available. Please contact Total Safety for more information.



Industrial Gas Detection Sensors

- Model 700
- "Environmentally Bullet-Proof"



Sensors utilize 6 distinctive gas detection technologies for harsh and extreme industrial applications that require monitoring of oxygen, toxic and flammable gases. Sensor design features an LED display, electropolished 316 stainless-steel housing, multi-layered transient spike protection circuitry and 100% encapsulated electronics. Sensor electronics are completely protected and immune to water ingress, vibration and corrosion.

Sensor modules come standard with dual redundant outputs, a linear 4-20 mA analog signal and a Modbus RS-485 serial output. Configuration and routine calibration are intuitive and menu-driven with fully scripted instructions.

2-year warranty for transmitter module. Sensor warranties vary based upon sensor type.

Explosion-proof. OSHA certified.

SENSORS AVAILABLE

Combustible Gas (Matched Pair Catalytic Bead)
Hydrogen Sulfide Gas (Solid State Metal Oxide Semiconductor)
Oxygen Gas (Galvanic Metal Air Battery)
Combustible Hydrocarbon and Carbon Dioxide Gas (Non-Dispersive Infrared Optical)
VOCs (PID)
Toxic Gas (Electrochemical Fuel Cell)

Contact Total Safety for the specific Detcon 700 Series Sensors or accessories needed.



MicroSafe™ Gas Detectors

- Models 500 and 600



Intelligent gas sensor assemblies designed to monitor ambient air for the presence of a wide variety of toxic and combustible gas compounds. MicroSafe™ intuitive software and non-intrusive operating interface enable menu-driven maintenance and programming through a glass lens cover using a hand-held magnet.

Sensors utilize universal transmitters allowing for complete interchangeability of sensors and transmitters. Sensors feature very low power consumption, explosion-proof instrument packages and intrinsically safe sensor housing. Calibration frequencies can vary according to the specific type of cell being used, but typical interval for these sensors ranges from 90 to 180 days.

Model 500 is universal and programmable for service over 40 gas detector sub-assemblies. Comes standard with a linear 4-20 mA output, one-man remote calibration, plug-in replaceable sensors, self-adjusting zero and span, LED indicators for fault and calibration status and a 16-character backlit alphanumeric display.

Model 600 is similar to the 500 Series but provides complete flexibility in system integration options with 3 alarm relays and an RS-485 serial communication port.

2-year warranty for control transmitter circuit. Sensor warranties vary based upon sensor type.

UL and CSA approved.

SENSORS AVAILABLE

Chemfet MOS Solid State H ₂ S
Catalytic Combustible Gas
Infrared Combustible Hydrocarbons and CO ₂
Electrochemical Toxic
O ₂ Deficiency
VOCs (PID)

Contact Total Safety for the specific Detcon 500 or 600 Series Sensors or accessories needed.



At Total Safety, you have access to an incredible inventory of up-to-date, up-to-service equipment for daily, weekly, monthly or custom term rental.



Single Sensor Control Card System

• Model 10 Series and 12 Series

Model 12 Series



Model 10 Series

Single sensor control cards can be integrated into one system to provide multi-function monitoring and control of a wide range of field devices such as gas detection sensors, flame and fire detection devices and process sensors. Each control card is capable of supervising a single field device, displaying the current status of that device and providing alarm relay outputs once user-configured limits are detected.

Gas detection control modules display the real-time concentration of a wide range

of toxic gases, combustible gases and oxygen sensors that output a 4-20 mA signal. Gas flame detection module displays real time status of any flame detector. An alarm annunciator module allows monitoring of smoke, thermal detectors, pull stations, and any dry contact closure device.

Series 10 features 2 active alarm relays, and Series 12 features 3 active alarm relays. Both models also have one fault condition relay each. Alarm relays can be user-configured as latching, non-latching, normally energized or normally de-energized via simple onboard programming jumpers. Front panel indicators include alarm status LEDs and a 7-segment LED digital display of the real time concentration of a field device when applicable. Each module is addressable via RS-485 serial communication, which can be connected to any DCS, PLC or SCADA host control device and can be used to drive optional remote Read-Only Displays.

Detcon manufactures a variety of pre-engineered NEMA 1 mainframe hardware assemblies including NEMA 4X weatherproof and NEMA 7X explosion-proof enclosures.

Comes with 5-year, fix-free service policy.

SENSOR CONTROL CARDS AVAILABLE

Combustible Gas (Catalytic and Infrared)
Hydrogen Sulfide Gas (Solid State MOS and Electrochemical)
Toxic Gas (Electrochemical and PID)
Oxygen Deficiency/Enrichment (Air Battery Electrochemical)

PROCESS CONTROLS OPTIONS

Pressure	Intrusion
Temperature	Thermal
Humidity	Smoke
Flow	Flame
Level	Level

SAFETY CONTROLS OPTIONS



Model X40 Integrated Alarm and Control Systems



- 8 - 32 channel capacity
- Backlit LCD screen display
- Analog or RS-485 input options
- Easy intuitive field configuration
- Wired or distributed I/O
- Extensive event datalogging
- SmartWireless® capable
- 2 packages available: Model X40-8 and Model X40-32

A low power alarm and control system designed to monitor multiple gas detection sensors and/or a wide range of other field devices. System is completely field programmable. Displays real time readings and field device status on a backlit LCD screen. During normal operation the screen displays alarm status and current reading for up to 8 field devices simultaneously, cycling through each of the active channel screens in sequential order.

Control system is designed to receive and supervise inputs using either 4-20mA DC or Serial RS-485 Modbus. Functions as a Modbus Master and can be customized and expanded based on individual application needs using stackable din-rail mounted I/O modules (contact Total Safety for more information). All I/O modules are individually addressable, can be mounted within the main system enclosure or installed remotely and operate on 11.5-30VDC. Wireless option can be used with RXT-320 SmartWireless® Transceivers. Contact Total Safety for more information. Enclosure options include NEMA 4X, NEMA 1 panel mount, and NEMA 7.

Model X40-8 provides power for up to eight field devices and can house up to six 4-channel I/O modules.

Model X40-32 provides power for up to 32 field devices and can house up to twelve 4-channel I/O modules.

1-year warranty.

Cat. No.	Description	System Capacity	Rental	Purchase Price
DETX408N4X	X40-8	Up to 8 field devices	YES	Call for Quote
DETX4032N1P	X40-32 with NEMA 1 panel mount enclosure	Up to 32 field devices	YES	Call for Quote
DETX4032N4X	X40-32 with NEMA 4X enclosure	Up to 32 field devices	YES	Call for Quote
DETX4032N7	X40-32 with NEMA 7 enclosure	Up to 32 field devices (analog or serial)	YES	Call for Quote

Contact Total Safety for specific system configuration needed.

Contact Total Safety for the specific Detcon Model 10 Series or 12 Series Sensors and components needed.



MCX-32 Integrated Alarm and Control Systems

- 64 channel capacity
- Large LCD with touch-screen user interface
- Analog, digital or Modbus RTU serial input capability
- Field expandable with auto configuration capability
- Versatile distributed I/O installation options
- Lightning and over-voltage protection standard
- Extensive event datalogging



A multi-channel integrated control system with a capacity of 64 active channels with the ability to receive and supervise inputs from a wide range of field devices with either a 4-20mA DC or serial output. Features a flexible design that allows the user to customize a system by choosing from a selection of individually addressable field devices or addressable input and output modules (contact Total Safety for more information). Four standard I/O modules are available: a 4-channel 4-20mA input module (DA-4), a 4-alarm relay output module (RL-4), a 4-channel 4-20mA output module (AO-4), and a 4-relay contact input module (DI-4). I/O modules are din-rail mounted and stackable allowing for seamless system expansion. The I/O modules may be mounted within the main system enclosure or in separate enclosures suitable for the area classification where installation will occur.

The real time status of each active sensor is displayed on a backlit LCD Touch Screen that also serves as a user interface. Display formats include field device function, tag name, current reading and alarm status. Each input can be configured for up to three alarm levels. A fourth alarm output is dedicated to fault diagnostics of field devices and internal operating systems. Each system application can include common and/or a limited number of discrete alarm relays. System accepts Type I or Type II compact flash cards for datalogging. Other features include one-touch alarm inhibit, alarm reset and alarm silence (acknowledge) functions. A 10/100 BaseT Ethernet port allows users to remotely monitor system from any PC or handheld device with an internet connection. An addressable RS-485 Modbus RTU slave port is available for communication with higher level automation and control systems such as PLCs, PCs and Distributed Control Systems.

The Model MCX-32 is available in 4 enclosure options: NEMA 1 panel mount, NEMA 1 rack mount, NEMA 4X epoxy fiberglass and NEMA 4X 316 Stainless Steel. The control system can be powered by 110-230 V AC or 24 V DC. Model MCX-32 Integrated Control Systems are intended for installation inside buildings or under cover avoiding direct sunlight.

All models include 110/220 V AC and 24 V DC power supply. Addressable I/O Modules are not included and are sold separately.

1-year warranty.

Cat. No.	Description	Rental	Purchase Price
DET95P323210000	NEMA 1 panel mount, 64 channel integrated control system in panel mount hardware assembly	YES	Call for Quote
DET95R323210000	NEMA 1 rack mount, 64 channel integrated control system in 19 in (48.3 cm) rack mount hardware assembly	YES	Call for Quote
DET954323210000	NEMA 4X epoxy fiberglass, 64 channel integrated control system in wall mount ABS weatherproof enclosure	YES	Call for Quote
DET954323213000	NEMA 4X 316 stainless steel, 64 channel integrated control system in 316 stainless steel wall mount weather-proof enclosure	YES	Call for Quote

Contact Total Safety for specific system configuration needed.



Model 880A/S Multichannel Alarm and Control System

- PLC based Modbus™ master control system
- Two addressable slave ports
- Distributed I/O options
- Simple menu driven programming
- 8 in (20.3 cm) color touch screen
- Front panel LED status indicators
- USB drive alarm event logging
- NEMA 1 and NEMA 4X packaging



Designed to serve as a host monitoring and control system for serial or analog networks of gas detection sensors, as well as a broad range of other field devices. Useful for applications requiring up to 32- or 72-channel devices. Up to 32 sensors can be configured for RS-485 Modbus. When using analog 4-20 mA input, capacity goes up to 72 devices. Can be programmed for any combination of gas detection functions including a wide range of toxic gases, combustible gas and oxygen deficiency.

Comes in a NEMA 4X surface mount or NEMA 1 panel mount enclosure. Features programmable discrete or zoned alarm relays, operator interface via color touch screen, front panel accessible plug-in PLC modules, 32-channel serial input device capacity, 72-channel analog device capacity, alarm history datalogging, front panel LED alarm function indicators and video color change for zone alarm conditions.

Remote read-only model can be installed up to 1000 ft from the main control unit and have the same display capability. Features color touch screen, addressable RS-422 communication, NEMA 1 and NEMA 4 version options, and standard or customizable screen.

1-year warranty.

SURFACE MOUNT CONTROL UNIT

Cat. No.	Description	Rental	Purchase Price
DET904880S00000	Serial Version		Call for Quote
DET904880A00000	Analog Version	YES	Call for Quote

PANEL MOUNT CONTROL UNIT

Cat. No.	Description	Rental	Purchase Price
DET901880S50005	Serial Version		Call for Quote
DET901880A50005	Analog Version	YES	Call for Quote

REMOTE DISPLAY

Cat. No.	Description	Purchase Price
DET904880R50000	Remote Display	Call for Quote

Contact Total Safety for specific system configuration needed.



GenII Touch-Screen Hybrid Monitor



Hybrid monitor is capable of receiving transmissions from both wired and WireFree sensor assemblies. Supports up to 32 WireFree sensor units, and up to four 4-20mA input sensors (when only 28 channels are setup as WireFree). Standard NEMA 4 enclosure.

Features a 7 in (17.8 cm) 800 x 480 pixel resistive touch-screen and is equipped with 4 dry contact 5-amp alarm relays. Relays may be latched or unlatched. Monitor is customizable with up to two radios: 900MHz, 900MHz (legacy), or 2.4GHz (for international use). These radio options allow the device to be backward compatible with GenI WireFree sensor assemblies, while also supporting GenII WireFree sensor units. Operates with 120/240V AC power-in and includes an RS-485 Modbus output.

Monitor is compatible with all Otis sensor assemblies, including those which house catalytic bead, infrared, PID and electrochemical sensors.

1-year limited warranty.

Cat. No.	Description	Rental	Purchase Price
OTIOI7032	GenII Touch-Screen Monitor	YES	Call for Quote



The H₂S Sensor from Otis Instruments

In 1983, Otis Instruments developed an exclusive metal oxide, solid state sensor that operates at an approximate surface temperature of 250°F (121°C), making the sensor ideal for use in a wide range of temperatures – from deserts to ice fields. This sensor, known as the OTIOI240 H₂S, is incorporated into the H₂S Solid State Sensor Assemblies on this page.



GenII Hybrid Monitor



Available with 12, 32 or 64 channels, supporting up to 12, 32 or 64 WireFree sensor units and up to four 4-20mA input sensors. Hybrid monitor is capable of receiving transmissions from both wired and WireFree™ sensor assemblies. Standard NEMA 4 enclosure.

Features two graphical 128 x 64 LCD displays and is equipped with 4

dry contact 5-amp alarm relays. Relays may be latched or unlatched. Monitor may be configured with either a 900MHz or 2.4GHz radio for backward compatibility with GenI WireFree sensor assemblies. Operates with either 12-35V DC or 110/240V AC power-in and includes an RS-485 Modbus output.

Monitor is compatible with all Otis sensor assemblies, including those which house catalytic bead, infrared, PID and electrochemical sensors.

1-year limited warranty.

Cat. No.	Description	Rental	Purchase Price
OTIOI7010	GenII Hybrid Monitor	YES	Call for Quote



GenII Explosion-Proof Monitor



A WireFree 32-channel explosion-proof monitor specifically designed for use in conjunction with Otis Instruments, Inc. GenII WireFree Sensor Assemblies. Monitor features 3 relays and is capable of monitoring up to 32 sensors.

The ability to mount the device near the entrance of a site allows the technician to immediately determine if it is safe to enter the site, making this device incredibly useful in any environment hosting hazardous gas.

Can be mounted as a traditional or self-contained system that requires 12-24 V DC or 120/240 V AC. Receives transmissions from WireFree sensors

via an on-board 900MHz or 2.4GHz GenII radio.

CSA certified for Class I, Division 1, Groups C and D.

1-year limited warranty.

Cat. No.	Description	Rental	Purchase Price
OTIOI7500	GenII Explosion-Proof Monitor	YES	Call for Quote



GenII WireFree Sensor Assembly



WireFree sensor assembly uses an electro-chemical sensor element to detect a variety of gases. Device comes standard with 102 x 64 graphical LCD screen, standard 3-button interface, custom explosion-proof enclosure, non-intrusive magnetic switches, and radio (choice of either 900MHz or 2.4GHz).

Features auto-setting Null, relay/alarm tests and battery voltage indication. One-person non-intrusive calibration/configuration is quick and easy: simply use a distributed magnet to activate the MENU, ADD and SUB buttons.

Sensor Assembly is designed to be self-contained and to last from 6-12 months on a single battery (time varies depending of amount of gas detected). Designed to reject EMI and other forms of interference in order to avoid false gas readings. Sensor functions by transmission of radio wave messages to the receiving controller every minute when there is no gas present, and every 5 sec when gas is present (and above the background gas level setting). Since each system's address is field adjustable, any OI-6900 sensor may be used as a replacement.

Electrical Classification: Class I, Division 1, Group C and D.

Safety Approvals: Zone 1 ATEX d IIB Ex d IIB

UL and CSA approved.

1-year limited warranty.

Cat. No.	Description	Rental	Purchase Price
OTIOI6900	GenII WireFree Sensor Assembly	YES	Call for Quote



GenII Sensor Assembly



A wired, WireFree or "Notis" sensor assembly that uses a catalytic bead, electro-chemical, PID or infrared sensor element to detect a variety of gases. Device comes standard with a 102 x 64 graphical LCD screen, standard 3-button interface, custom explosion-proof enclosure, and magnetic switches for non-intrusive calibration.

System is designed to monitor gas in hostile environments without the use of wires or conduit from the controller to the sensor. Features 4-20mA (3-wire) or RS-485 Modbus output, and can be customized to offer relay and radio (900MHz or 2.4GHz) options. Device has been designed to reject EMI and other forms of interference in order to avoid false gas readings.

Electrical Classification: Class I, Division 1, Group C and D.

Safety Approvals: Zone 1 ATEX d IIB Ex d IIB

UL and CSA approved.

1-year limited warranty.

Cat. No.	Description	Rental	Purchase Price
OTIOI6000	GenII Sensor Assembly	YES	Call for Quote



ProSafe 4X4 Four-Channel Gas Monitor

The 4-channel gas monitor functions as a transmission controller. Monitors toxic, combustibles and oxygen gas levels in ambient air by receiving and displaying 4-20mA signals from wired sensor assemblies (sold separately). Contact Total Safety for a complete list of available sensor assemblies.

Features a 4-digit display with a NEMA 4 enclosure with a translucent door and is equipped with 4 independent 10-amp alarm relays that are full-scale adjustable: 2 DC-only relays and 2 dry contact relays. Operates with either 12-24V DC or 11V AC power-in.

1-year limited warranty.

Cat. No.	Description	Rental	Purchase Price
OTIOI744AC	Monitor with 110V AC and 12-24V DC	YES	Call for Quote
OTIOI744DC	Monitor with 12-24V DC only	YES	Call for Quote



**GenII ProSafe 4 x 4² Monitor**

4 channel gas monitor functions as a transmission controller. Monitors toxic, combustible and oxygen gas levels in ambient air by receiving and displaying 4-20mA signals from up to 4 wired sensor assemblies.

Features a 160 x 104 graphical LCD display and is equipped with 4 "dry contact" (Form C) 5-amp alarm relays (with 4-amp fuses) that are full-scale adjustable. Operates with either 12-35 Volts DC or 110/240 Volts AC power-in and includes a Modbus output with optional real-time monitoring software.

1-year limited warranty.

Cat. No.	Description	Rental	Purchase Price
OTIOI7440DC	Monitor with DC Power Only	YES	Call for Quote
OTIOI7440AC	Monitor with AC or DC Power	YES	Call for Quote

**GenII ProSafe 4x32² Monitor**

The 32-channel gas monitor functions as a transmission controller. Monitors toxic, combustible and oxygen gas levels in ambient air by receiving and displaying RS-485 Modbus signals from up to 32 sensor assemblies.

Features a 160 x 104 graphical LCD display and is equipped with 4 "dry contact" (Form C) 5-amp alarm relays (with 4-amp fuses) that are full-scale adjustable. Operates with either 12-35 Volts DC or 110/240 Volts AC power-in and includes a Modbus output with optional real-time monitoring software.

1-year limited warranty.

Cat. No.	Description	Rental	Purchase Price
OTIOI7432DC	Monitor with DC Power Only	YES	Call for Quote
OTIOI7432AC	Monitor with AC or DC Power	YES	Call for Quote

Hydrogen Sulfide Specifications

Hydrogen Sulfide Specifications

Formula: H₂S

CAS 7783-06-4

Synonyms & Trade Names: Hydrosulfuric Acid, Sewer Gas, Sulfretted Hydrogen

Exposure Limits:

NIOSH REL: 10 ppm (15 mg/m³) [8 hours]

OSHA PEL: 20 ppm, 50 ppm [10-minute maximum peak]

IDLH: 100 ppm

Conversion: 1 ppm = 1.40 mg/m³

Physical Description: Colorless gas with a strong odor of rotten eggs

MW: 34.1 BP: -77°F (-61°C) FRZ: -122°F (-86°C) RGasD: 1.19

VP: 17.6 atm IE (IP): 10.46 eV FL.P: NA (Gas) Sol: 0.4%

LEL: 4.0% UEL: 44.0% Flammable Gas

Incompatibilities & Reactivities: strong oxidizers, strong nitric acid, metals

Measurement Methods: NIOSH 6013, OSHA ID141

Otis Instruments Sensors

Otis Instruments offers a complete line of catalytic bead, infrared, solid state and electrochemical sensors in compatible sensor assemblies. Customizable for a variety of gases, these sensors have given many industry experts reason to refer to Otis Instruments as one of the world's foremost authorities in gas detection.

Contact Total Safety for more information on Otis Instruments Sensors and Sensor Assemblies.



PT2008 Wireless Gas Monitor and Controller



Monitor is capable of receiving transmissions from up to 24 Ultra 1000 series sensors over a wireless network. NEMA 4X enclosure.

Features a graphical LCD display with backlight and is equipped with 6 Form C type 5-amp alarm relays. Monitor uses license-free frequencies, 900 MHz (North America) or 2.4 GHz (worldwide). Operates with either 24V DC or 120/240V AC power-in and includes an RS-485 MODBUS output. Other features include optional cell modem for sending Alarm Text Messages using cellular network and alarm/event log with date and time stamp. Event log is capable of storing up to 300 event log records.

Cat. No.	Description	Rental	Purchase Price
PEMPT2008	Wireless Gas Monitor and Controller	YES	Call for quote



Ultra 1000 Wireless Sensor

- Solar powered stand-alone sensor with integrated battery charger



Sensor assembly detects a variety of toxic and combustible gases. Features 2 x 8 LCD screen, explosion-proof enclosure with 316 stainless steel detector head assembly, and easy menu driven singlepoint, single-person calibration/configuration. Uses 900MHz and 2.4GHz ISM bands.

Solar powered with internal NiMH battery and integrated battery charger. Sensor is capable of auto-detecting network; units come online automatically. Complete data communication with PT2008 Wireless Monitor, including gas reading and diagnostic data.

Electrical Classification: Class I, Division 1, Group C and D.

1-year limited warranty.

SENSORS AVAILABLE

Gas	Range
Hydrogen Sulfide	100 ppm
Carbon Monoxide	100 ppm/1000 ppm
Oxygen	25%
Sulfur Dioxide	50 ppm
Combustibles	0 to 100 % LEL

Contact Total Safety for a complete listing of available sensors.



Total Safety's world class gas detection solutions include the rental or purchase of:

- Single gas monitors
- Multigas monitors
- Photoionization detectors (PIDs)
- Flameionization detectors (FIDs)
- Air sampling pumps and monitors
- Portable and fixed area monitoring systems

**Eagle Quantum Premier®
(EQP) System**

A configurable, distributed, intelligent safety system that is flexible enough to provide flame and/or gas detection with alarm signaling, notification, extinguishing agent release, and/or deluge operation.

Ideally suited for harsh industrial applications, the system features extended functionality such as addressable smoke and heat detection for living quarters and control rooms; device calibration data and event logging; programmable logic; editable logic diagrams resident in the controller; optional controller operation redundancy; extensive advanced diagnostics and Safety System Solutions (S3) software. System components (such as the following) are integrated on a reliable, fault-tolerant digital communication network.

Certificates and Approvals:

- FM certified to ANSI/NFPA 72: 2010 and ANSI/NFPA 33
- FM certified to combustible and low-concentration ppm gas detection
- Loss Prevention Certification Board (LPCB) approval to EN54 and EN12094
- Multiple hazardous locations certifications
- Exida certified SIL-2
- US Coast Guard type approval
- Lloyds Register of Shipping type approval

**X3301 Multispectrum IR
Flame Detector**

Uses signal processing algorithms supported by an embedded 32-bit microprocessor to provide continuous protection in the presence of false alarm sources and environments with infrared radiation present.

Detector is suitable for outdoor and indoor applications that require the highest level of false alarm rejection and fire detection performance. Available in aluminum or 316 stainless steel for installation in the harshest environments.

Features a detection range to n-Heptane of 210 ft (64.0 m), and a solid cone of vision for methane. Additional features include standard fire alarm, fault and auxiliary relays with an isolated 0 to 20mA output model with optional HART communication.

SIL-2 rated. Heated optics for moisture and ice resistance. Multiple sensitivity levels for variety of applications.

5-year warranty.

SYSTEM COMPONENTS AVAILABLE

Controller	The EQ3001 Eagle Quantum Premier (EQP) Fire and Gas Detection/Releasing System serves multiple roles in the monitoring and protection of hazardous areas. A third-party certified SIL 2 option is available.
Safety System Software	EQP Safety System Software (S3) provides a full featured data acquisition, display, and configuration package.
Flame Detectors	System offers flame detection options for a myriad of industrial applications. All flame detectors have been tested to global performance approvals.
Gas Detectors	Full line of hazardous gas detection configurations range from a variety of fixed gas combustible and toxic gas detectors to a complete line of display transmitters, gas controllers, and gas control panels.
Inputs and Outputs	An EQP system can include a wide variety of field input and output (i/o) types, which are brought into the EQP system by Det-Tronics input/output modules.

Contact Total Safety for specific system configuration needed.